

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250287043

Kind Code

A1

Publication Date

September 11, 2025

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ENCODER, DECODER, ENCODING METHOD, AND DECODING METHOD

Abstract

An encoder includes circuitry and memory coupled to the circuitry. In operation, the circuitry: writes, into a sequence parameter set which is header information of a sequence, video usability information syntax which is information for realizing an additional function in display of an image, and syntax different from the video usability information syntax. The syntax includes at least one parameter related to display timing of the image.

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Family ID: 1000008620882

Appl. No.: 19/217014

Filed: May 23, 2025

Related U.S. Application Data

parent US continuation 18745003 20240617 parent-grant-document US 12348784 child US 19217014

parent US continuation 18133070 20230411 parent-grant-document US 12052442 child US 18745003

parent US continuation 17379302 20210719 parent-grant-document US 11665370 child US 18133070

parent WO continuation PCT/JP2020/007167 20200221 PENDING child US 17379302

us-provisional-application US 62808528 20190221

Publication Classification

**Int. Cl.: H04N19/70 (20140101); H04N19/169 (20140101); H04N19/423 (20140101);
H04N19/46 (20140101)**

U.S. Cl.:

**CPC H04N19/70 (20141101); H04N19/188 (20141101); H04N19/423 (20141101);
H04N19/46 (20141101);**

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/745,003 filed on Jun. 17, 2024, which is a continuation of U.S. application Ser. No. 18/133,070, now U.S. Pat. No. 12,052,442, filed on Apr. 11, 2023, which is a continuation of U.S. application Ser. No. 17/379,302, now U.S. Pat. No. 11,665,370, filed on Jul. 19, 2021, which is a continuation application of PCT International Patent Application Number PCT/JP2020/007167 filed on Feb. 21, 2020, claiming the benefit of priority of U.S. Provisional Application No. 62/808,528 filed on Feb. 21, 2019. The entire disclosures of the above-identified applications, including the specifications, drawings, and claims are incorporated herein by reference in their entirety.

1. TECHNICAL FIELD

[0002] This disclosure relates to video coding, and particularly to video encoding and decoding systems, components, and methods.

2. DESCRIPTION OF THE RELATED ART

[0003] With advancement in video coding technology, from H.261 and MPEG-1 to H.264/AVC (Advanced Video Coding), MPEG-LA, H.265/HEVC (High Efficiency Video Coding) and H.266/VVC (Versatile Video Codec), there remains a constant need to provide improvements and optimizations to the video coding technology to process an ever-increasing amount of digital video data in various applications.

[0004] H.265 (ISO/IEC 23008-2/HEVC (High Efficiency Video Coding)) relates to an example of a conventional Standard regarding the above-described video coding technology.

SUMMARY

[0005] For example, an encoder according to an aspect of the present disclosure includes: circuitry, and memory coupled to the circuitry, wherein in operation, the circuitry: writes, into a sequence parameter set which is header information of a sequence, video usability information syntax which is information for realizing an additional function in display of an image, and syntax different from the video usability information syntax, and the syntax includes at least one parameter related to display timing of the image.

[0006] Some implementations of embodiments of the present disclosure may improve an encoding efficiency, may simply be an encoding/decoding process, may accelerate an encoding/decoding process speed, may efficiently select appropriate components/operations used in encoding and decoding such as appropriate filter, block size, motion vector, reference picture, reference block, etc.

[0007] Additional benefits and advantages of the disclosed embodiments will become apparent from the specification and drawings. The benefits and/or advantages may be individually obtained by the various embodiments and features of the specification and drawings, not all of which need to be provided in order to obtain one or more of such benefits and/or advantages.

[0008] It is to be noted that general or specific embodiments may be implemented as a system, a

method, an integrated circuit, a computer program, a storage medium, or any selective combination thereof.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] These and other objects, advantages and features of the disclosure will become apparent from the following description thereof taken in conjunction with the accompanying drawings that illustrate a specific embodiment of the present disclosure.

[0010] FIG. 1 is a block diagram illustrating a functional configuration of an encoder according to an embodiment;

[0011] FIG. 2 is a flow chart indicating one example of an overall encoding process performed by the encoder;

[0012] FIG. 3 is a conceptual diagram illustrating one example of block splitting;

[0013] FIG. 4A is a conceptual diagram illustrating one example of a slice configuration;

[0014] FIG. 4B is a conceptual diagram illustrating one example of a tile configuration;

[0015] FIG. 5A is a chart indicating transform basis functions for various transform types;

[0016] FIG. 5B is a conceptual diagram illustrating example spatially varying transforms (SVT);

[0017] FIG. 6A is a conceptual diagram illustrating one example of a filter shape used in an adaptive loop filter (ALF);

[0018] FIG. 6B is a conceptual diagram illustrating another example of a filter shape used in an ALF;

[0019] FIG. 6C is a conceptual diagram illustrating another example of a filter shape used in an ALF;

[0020] FIG. 7 is a block diagram indicating one example of a specific configuration of a loop filter which functions as a deblocking filter (DBF);

[0021] FIG. 8 is a conceptual diagram indicating an example of a deblocking filter having a symmetrical filtering characteristic with respect to a block boundary;

[0022] FIG. 9 is a conceptual diagram for illustrating a block boundary on which a deblocking filter process is performed;

[0023] FIG. 10 is a conceptual diagram indicating examples of Bs values;

[0024] FIG. 11 is a flow chart illustrating one example of a process performed by a prediction processor of the encoder;

[0025] FIG. 12 is a flow chart illustrating another example of a process performed by the prediction processor of the encoder;

[0026] FIG. 13 is a flow chart illustrating another example of a process performed by the prediction processor of the encoder;

[0027] FIG. 14 is a conceptual diagram illustrating sixty-seven intra prediction modes used in intra prediction in an embodiment;

[0028] FIG. 15 is a flow chart illustrating an example basic processing flow of inter prediction;

[0029] FIG. 16 is a flow chart illustrating one example of derivation of motion vectors;

[0030] FIG. 17 is a flow chart illustrating another example of derivation of motion vectors;

[0031] FIG. 18 is a flow chart illustrating another example of derivation of motion vectors;

[0032] FIG. 19 is a flow chart illustrating an example of inter prediction in normal inter mode;

[0033] FIG. 20 is a flow chart illustrating an example of inter prediction in merge mode;

[0034] FIG. 21 is a conceptual diagram for illustrating one example of a motion vector derivation process in merge mode;

[0035] FIG. 22 is a flow chart illustrating one example of frame rate up conversion (FRUC) process;

[0036] FIG. **23** is a conceptual diagram for illustrating one example of pattern matching (bilateral matching) between two blocks along a motion trajectory;

[0037] FIG. **24** is a conceptual diagram for illustrating one example of pattern matching (template matching) between a template in a current picture and a block in a reference picture;

[0038] FIG. **25A** is a conceptual diagram for illustrating one example of deriving a motion vector of each sub-block based on motion vectors of a plurality of neighboring blocks;

[0039] FIG. **25B** is a conceptual diagram for illustrating one example of deriving a motion vector of each sub-block in affine mode in which three control points are used;

[0040] FIG. **26A** is a conceptual diagram for illustrating an affine merge mode;

[0041] FIG. **26B** is a conceptual diagram for illustrating an affine merge mode in which two control points are used;

[0042] FIG. **26C** is a conceptual diagram for illustrating an affine merge mode in which three control points are used;

[0043] FIG. **27** is a flow chart illustrating one example of a process in affine merge mode;

[0044] FIG. **28A** is a conceptual diagram for illustrating an affine inter mode in which two control points are used;

[0045] FIG. **28B** is a conceptual diagram for illustrating an affine inter mode in which three control points are used;

[0046] FIG. **29** is a flow chart illustrating one example of a process in affine inter mode;

[0047] FIG. **30A** is a conceptual diagram for illustrating an affine inter mode in which a current block has three control points and a neighboring block has two control points;

[0048] FIG. **30B** is a conceptual diagram for illustrating an affine inter mode in which a current block has two control points and a neighboring block has three control points;

[0049] FIG. **31A** is a flow chart illustrating a merge mode process including decoder motion vector refinement (DMVR);

[0050] FIG. **31B** is a conceptual diagram for illustrating one example of a DMVR process;

[0051] FIG. **32** is a flow chart illustrating one example of generation of a prediction image;

[0052] FIG. **33** is a flow chart illustrating another example of generation of a prediction image;

[0053] FIG. **34** is a flow chart illustrating another example of generation of a prediction image;

[0054] FIG. **35** is a flow chart illustrating one example of a prediction image correction process performed by an overlapped block motion compensation (OBMC) process;

[0055] FIG. **36** is a conceptual diagram for illustrating one example of a prediction image correction process performed by an OBMC process;

[0056] FIG. **37** is a conceptual diagram for illustrating generation of two triangular prediction images;

[0057] FIG. **38** is a conceptual diagram for illustrating a model assuming uniform linear motion;

[0058] FIG. **39** is a conceptual diagram for illustrating one example of a prediction image generation method using a luminance correction process performed by a local illumination compensation (LIC) process;

[0059] FIG. **40** is a block diagram illustrating a mounting example of the encoder;

[0060] FIG. **41** is a block diagram illustrating a functional configuration of a decoder according to an embodiment;

[0061] FIG. **42** is a flow chart illustrating one example of an overall decoding process performed by the decoder;

[0062] FIG. **43** is a flow chart illustrating one example of a process performed by a prediction processor of the decoder;

[0063] FIG. **44** is a flow chart illustrating another example of a process performed by the prediction processor of the decoder;

[0064] FIG. **45** is a flow chart illustrating an example of inter prediction in normal inter mode in the decoder;

[0065] FIG. **46** is a block diagram illustrating a mounting example of the decoder;
[0066] FIG. **47** is a diagram illustrating an example of an essential display information (EDI) syntax structure;
[0067] FIG. **48** is a diagram illustrating an example of a video usability information (VUI) syntax structure;
[0068] FIG. **49** is a diagram illustrating an example of EDI and VUI syntax included in a sequence parameter set;
[0069] FIG. **50** is a flowchart illustrating an example of a process of reading EDI information and VUI information by a receiver that receives a bitstream conforming to Aspect 1;
[0070] FIG. **51** is a flowchart illustrating an example of an operation of a receiver that receives a bitstream conforming to Aspect 1;
[0071] FIG. **52** is a flowchart illustrating another example of an operation of a receiver that receives a bitstream conforming to Aspect 1;
[0072] FIG. **53A** is a first diagram illustrating an example of an EDI syntax structure;
[0073] FIG. **53B** is a second diagram illustrating an example of an EDI syntax structure;
[0074] FIG. **54A** is a first diagram illustrating an example of syntax conforming to Aspect 2;
[0075] FIG. **54B** is a second diagram illustrating an example of syntax conforming to Aspect 2;
[0076] FIG. **55** is a flow chart illustrating an operation performed by the encoder;
[0077] FIG. **56** is a flow chart illustrating an operation performed by the decoder;
[0078] FIG. **57** is a block diagram illustrating an overall configuration of a content providing system for implementing a content distribution service;
[0079] FIG. **58** is a conceptual diagram illustrating one example of an encoding structure in scalable encoding;
[0080] FIG. **59** is a conceptual diagram illustrating one example of an encoding structure in scalable encoding;
[0081] FIG. **60** is a conceptual diagram illustrating an example of a display screen of a web page;
[0082] FIG. **61** is a conceptual diagram illustrating an example of a display screen of a web page;
[0083] FIG. **62** is a block diagram illustrating one example of a smartphone; and
[0084] FIG. **63** is a block diagram illustrating an example of a configuration of a smartphone.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0085] An encoder according to an aspect of the present disclosure includes circuitry, and memory coupled to the circuitry. In operation, the circuitry writes, into a bitstream, a flag indicating whether first display information which includes information related to image aspect ratio, information related to color description, and information related to chroma location is indicated in the bitstream.

[0086] Accordingly, when the encoder writes, into the bitstream, a flag indicating that first display information is not indicated in the bitstream, the need to search for the first display information within the bitstream is eliminated, and thus the processing amount is reduced. Therefore, the encoding efficiency of the encoder improves.

[0087] For example, the first display information may include the information related to image aspect ratio, the information related to color description, and the information related to chroma location, in stated order.

[0088] Accordingly, the information related to image aspect ratio, the information related to color description, and the information related to chroma location are written into the bitstream in the order that they are included in the first display information. As a result, the encoder does not need to search for these items of information one after the other within the bitstream, and thus the processing amount is reduced. Therefore, the encoding efficiency of the encoder improves.

[0089] For example, the information related to image aspect ratio may include a flag indicating whether the information related to image aspect ratio is to be encoded, the information related to color description may include a flag indicating whether the information related to color description is to be encoded, and the information related to chroma location may include a flag indicating

whether the information related to chroma location is to be encoded.

[0090] Accordingly, the encoder can encode the respective information according to the flag included in each of the information. As a result, all of the information need not necessarily be encoded, and thus the processing amount is reduced. Therefore, the encoding efficiency of the encoder improves.

[0091] For example, the circuitry may write, into the bitstream, a flag indicating whether second display information different from the first display information is indicated in the bitstream.

[0092] Accordingly, when the encoder writes, into the bitstream, a flag indicating that second display information is not indicated in the bitstream, the need to search for the first display information within the bitstream is eliminated, and thus the processing amount is reduced.

Therefore, the encoding efficiency of the encoder improves.

[0093] A decoder according to an aspect of the present disclosure includes circuitry, and memory coupled to the circuitry. In operation, the circuitry reads, from a bitstream, a flag indicating whether first display information which includes information related to image aspect ratio, information related to color description, and information related to chroma location is indicated in the bitstream.

[0094] Accordingly, when the decoder reads, from the bitstream, a flag indicating that first display information is not indicated in the bitstream, the need to search for the first display information within the bitstream is eliminated, and thus the processing amount is reduced. Therefore, the processing efficiency of the decoder improves.

[0095] For example, the first display information may include the information related to image aspect ratio, the information related to color description, and the information related to chroma location, in stated order.

[0096] Accordingly, the information related to image aspect ratio, the information related to color description, and the information related to chroma location are read from the bitstream in the order that they are included in the first display information. As a result, the decoder does not need to search for these items of information one after the other within the bitstream, the processing amount is reduced. Therefore, the processing efficiency of the decoder improves.

[0097] For example, the information related to image aspect ratio may include a flag indicating whether the information related to image aspect ratio is to be encoded, the information related to color description may include a flag indicating whether the information related to color description is to be encoded, and the information related to chroma location may include a flag indicating whether the information related to chroma location is to be encoded.

[0098] Accordingly, the decoder can decode the respective information according to the flag included in each of the above-described information. As a result, all of the information need not necessarily be decoded, and thus the processing amount is reduced. Therefore, the processing efficiency of the decoder improves.

[0099] For example, the circuitry may read, from the bitstream, a flag indicating whether second display information different from the first display information is indicated in the bitstream.

[0100] Accordingly, when the decoder reads, from the bitstream, a flag indicating that second display information is not indicated in the bitstream, the need to search for the second display information within the bitstream is eliminated, and thus the processing amount is reduced.

Therefore, the processing efficiency of the decoder improves.

[0101] An encoding method according to an aspect of the present disclosure includes: writing, into a bitstream, a flag indicating whether first display information which includes information related to image aspect ratio, information related to color description, and information related to chroma location is indicated in the bitstream.

[0102] Accordingly, in the encoding method, when a flag indicating that second display information is not indicated in the bitstream is written in the bitstream, the need to search for the first display information within the bitstream is eliminated, and thus the processing amount is reduced. Therefore, the encoding method improves encoding efficiency.

[0103] A decoding method according to an aspect of the present disclosure includes: reading, from a bitstream, a flag indicating whether first display information which includes information related to image aspect ratio, information related to color description, and information related to chroma location is indicated in the bitstream.

[0104] Accordingly, in the decoding method, when a flag indicating that first display information is not indicated in the bitstream is read from the bitstream, the need to search for the first display information within the bitstream is eliminated, and thus the processing amount is reduced. Therefore, the decoding method improves the processing efficiency.

[0105] Furthermore, these general and specific aspects may be implemented using a system, a device, a method, an integrated circuit, a computer program, a non-transient recording medium such as a computer-readable CD-ROM, or any combination of systems, devices, methods, integrated circuits, computer programs or recording media.

[0106] Hereinafter, embodiments will be described with reference to the drawings. Note that the embodiments described below each show a general or specific example. The numerical values, shapes, materials, components, the arrangement and connection of the components, steps, the relation and order of the steps, etc., indicated in the following embodiments are mere examples, and are not intended to limit the scope of the claims.

[0107] Embodiments of an encoder and a decoder will be described below. The embodiments are examples of an encoder and a decoder to which the processes and/or configurations presented in the description of aspects of the present disclosure are applicable. The processes and/or configurations can also be implemented in an encoder and a decoder different from those according to the embodiments. For example, regarding the processes and/or configurations as applied to the embodiments, any of the following may be implemented:

[0108] (1) Any of the components of the encoder or the decoder according to the embodiments presented in the description of aspects of the present disclosure may be substituted or combined with another component presented anywhere in the description of aspects of the present disclosure.

[0109] (2) In the encoder or the decoder according to the embodiments, discretionary changes may be made to functions or processes performed by one or more components of the encoder or the decoder, such as addition, substitution, removal, etc., of the functions or processes. For example, any function or process may be substituted or combined with another function or process presented anywhere in the description of aspects of the present disclosure.

[0110] (3) In methods implemented by the encoder or the decoder according to the embodiments, discretionary changes may be made such as addition, substitution, and removal of one or more of the processes included in the method. For example, any process in the method may be substituted or combined with another process presented anywhere in the description of aspects of the present disclosure.

[0111] (4) One or more components included in the encoder or the decoder according to embodiments may be combined with a component presented anywhere in the description of aspects of the present disclosure, may be combined with a component including one or more functions presented anywhere in the description of aspects of the present disclosure, and may be combined with a component that implements one or more processes implemented by a component presented in the description of aspects of the present disclosure.

[0112] (5) A component including one or more functions of the encoder or the decoder according to the embodiments, or a component that implements one or more processes of the encoder or the decoder according to the embodiments, may be combined or substituted with a component presented anywhere in the description of aspects of the present disclosure, with a component including one or more functions presented anywhere in the description of aspects of the present disclosure, or with a component that implements one or more processes presented anywhere in the description of aspects of the present disclosure.

[0113] (6) In methods implemented by the encoder or the decoder according to the embodiments,

any of the processes included in the method may be substituted or combined with a process presented anywhere in the description of aspects of the present disclosure or with any corresponding or equivalent process.

[0114] (7) One or more processes included in methods implemented by the encoder or the decoder according to the embodiments may be combined with a process presented anywhere in the description of aspects of the present disclosure.

[0115] (8) The implementation of the processes and/or configurations presented in the description of aspects of the present disclosure is not limited to the encoder or the decoder according to the embodiments. For example, the processes and/or configurations may be implemented in a device used for a purpose different from the moving picture encoder or the moving picture decoder disclosed in the embodiments.

[Encoder]

[0116] First, an encoder according to an embodiment will be described. FIG. 1 is a block diagram illustrating a functional configuration of encoder **100** according to the embodiment. Encoder **100** is a video encoder which encodes a video in units of a block.

[0117] As illustrated in FIG. 1, encoder **100** is an apparatus which encodes an image in units of a block, and includes splitter **102**, subtractor **104**, transformer **106**, quantizer **108**, entropy encoder **110**, inverse quantizer **112**, inverse transformer **114**, adder **116**, block memory **118**, loop filter **120**, frame memory **122**, intra predictor **124**, inter predictor **126**, and prediction controller **128**.

[0118] Encoder **100** is implemented as, for example, a generic processor and memory. In this case, when a software program stored in the memory is executed by the processor, the processor functions as splitter **102**, subtractor **104**, transformer **106**, quantizer **108**, entropy encoder **110**, inverse quantizer **112**, inverse transformer **114**, adder **116**, loop filter **120**, intra predictor **124**, inter predictor **126**, and prediction controller **128**. Alternatively, encoder **100** may be implemented as one or more dedicated electronic circuits corresponding to splitter **102**, subtractor **104**, transformer **106**, quantizer **108**, entropy encoder **110**, inverse quantizer **112**, inverse transformer **114**, adder **116**, loop filter **120**, intra predictor **124**, inter predictor **126**, and prediction controller **128**.

[0119] Hereinafter, an overall flow of processes performed by encoder **100** is described, and then each of constituent elements included in encoder **100** will be described.

[Overall Flow of Encoding Process]

[0120] FIG. 2 is a flow chart indicating one example of an overall encoding process performed by encoder **100**.

[0121] First, splitter **102** of encoder **100** splits each of pictures included in an input image which is a video into a plurality of blocks having a fixed size (e.g., 128×128 pixels) (Step Sa_1). Splitter **102** then selects a splitting pattern for the fixed-size block (also referred to as a block shape) (Step Sa_2). In other words, splitter **102** further splits the fixed-size block into a plurality of blocks which form the selected splitting pattern. Encoder **100** performs, for each of the plurality of blocks, Steps Sa_3 to Sa_9 for the block (that is a current block to be encoded).

[0122] In other words, a prediction processor which includes all or part of intra predictor **124**, inter predictor **126**, and prediction controller **128** generates a prediction signal (also referred to as a prediction block) of the current block to be encoded (also referred to as a current block) (Step Sa_3).

[0123] Next, subtractor **104** generates a difference between the current block and a prediction block as a prediction residual (also referred to as a difference block) (Step Sa_4).

[0124] Next, transformer **106** transforms the difference block and quantizer **108** quantizes the result, to generate a plurality of quantized coefficients (Step Sa_5). It is to be noted that the block having the plurality of quantized coefficients is also referred to as a coefficient block.

[0125] Next, entropy encoder **110** encodes (specifically, entropy encodes) the coefficient block and a prediction parameter related to generation of a prediction signal to generate an encoded signal (Step Sa_6). It is to be noted that the encoded signal is also referred to as an encoded bitstream, a

compressed bitstream, or a stream.

[0126] Next, inverse quantizer **112** performs inverse quantization of the coefficient block and inverse transformer **114** performs inverse transform of the result, to restore a plurality of prediction residuals (that is, a difference block) (Step Sa_7).

[0127] Next, adder **116** adds the prediction block to the restored difference block to reconstruct the current block as a reconstructed image (also referred to as a reconstructed block or a decoded image block) (Step Sa_8). In this way, the reconstructed image is generated.

[0128] When the reconstructed image is generated, loop filter **120** performs filtering of the reconstructed image as necessary (Step Sa_9).

[0129] Encoder **100** then determines whether encoding of the entire picture has been finished (Step Sa_10). When determining that the encoding has not yet been finished (No in Step Sa_10), processes from Step Sa_2 are executed repeatedly.

[0130] Although encoder **100** selects one splitting pattern for a fixed-size block, and encodes each block according to the splitting pattern in the above-described example, it is to be noted that each block may be encoded according to a corresponding one of a plurality of splitting patterns. In this case, encoder **100** may evaluate a cost for each of the plurality of splitting patterns, and, for example, may select the encoded signal obtainable by encoding according to the splitting pattern which yields the smallest cost as an encoded signal which is output.

[0131] As illustrated, the processes in Steps Sa_1 to Sa_10 are performed sequentially by encoder **100**. Alternatively, two or more of the processes may be performed in parallel, the processes may be reordered, etc.

[Splitter]

[0132] Splitter **102** splits each of pictures included in an input video into a plurality of blocks, and outputs each block to subtractor **104**. For example, splitter **102** first splits a picture into blocks of a fixed size (for example, 128×128). Other fixed block sizes may be employed. The fixed-size block is also referred to as a coding tree unit (CTU). Splitter **102** then splits each fixed-size block into blocks of variable sizes (for example, 64×64 or smaller), based on recursive quadtree and/or binary tree block splitting. In other words, splitter **102** selects a splitting pattern. The variable-size block is also referred to as a coding unit (CU), a prediction unit (PU), or a transform unit (TU). It is to be noted that, in various kinds of processing examples, there is no need to differentiate between CU, PU, and TU; all or some of the blocks in a picture may be processed in units of a CU, a PU, or a TU.

[0133] FIG. 3 is a conceptual diagram illustrating one example of block splitting according to an embodiment. In FIG. 3, the solid lines represent block boundaries of blocks split by quadtree block splitting, and the dashed lines represent block boundaries of blocks split by binary tree block splitting.

[0134] Here, block **10** is a square block having 128×128 pixels (128×128 block). This 128×128 block **10** is first split into four square 64×64 blocks (quadtree block splitting).

[0135] The upper-left 64×64 block is further vertically split into two rectangular 32×64 blocks, and the left 32×64 block is further vertically split into two rectangular 16×64 blocks (binary tree block splitting). As a result, the upper-left 64×64 block is split into two 16×64 blocks **11** and **12** and one 32×64 block **13**.

[0136] The upper-right 64×64 block is horizontally split into two rectangular 64×32 blocks **14** and **15** (binary tree block splitting).

[0137] The lower-left 64×64 block is first split into four square 32×32 blocks (quadtree block splitting). The upper-left block and the lower-right block among the four 32×32 blocks are further split. The upper-left 32×32 block is vertically split into two rectangle 16×32 blocks, and the right 16×32 block is further horizontally split into two 16×16 blocks (binary tree block splitting). The lower-right 32×32 block is horizontally split into two 32×16 blocks (binary tree block splitting). As a result, the lower-left 64×64 block is split into 16×32 block **16**, two 16×16 blocks **17** and **18**, two

32×32 blocks **19** and **20**, and two 32×16 blocks **21** and **22**.

[0138] The lower-right 64×64 block **23** is not split.

[0139] As described above, in FIG. 3, block **10** is split into thirteen variable-size blocks **11** through **23** based on recursive quadtree and binary tree block splitting. This type of splitting is also referred to as quadtree plus binary tree (QTBT) splitting.

[0140] It is to be noted that, in FIG. 3, one block is split into four or two blocks (quadtree or binary tree block splitting), but splitting is not limited to these examples. For example, one block may be split into three blocks (ternary block splitting). Splitting including such ternary block splitting is also referred to as multi-type tree (MBT) splitting.

[Picture Structure: Slice/Tile]

[0141] A picture may be configured in units of one or more slices or tiles in order to decode the picture in parallel. The picture configured in units of one or more slices or tiles may be configured by splitter **102**.

[0142] Slices are basic encoding units included in a picture. A picture may include, for example, one or more slices. In addition, a slice includes one or more successive coding tree units (CTU).

[0143] FIG. 4A is a conceptual diagram illustrating one example of a slice configuration. For example, a picture includes 11×8 CTUs and is split into four slices (slices 1 to 4). Slice 1 includes sixteen CTUs, slice 2 includes twenty-one CTUs, slice 3 includes twenty-nine CTUs, and slice 4 includes twenty-two CTUs. Here, each CTU in the picture belongs to one of the slices. The shape of each slice is a shape obtainable by splitting the picture horizontally. A boundary of each slice does not need to coincide with an image end, and may coincide with any of the boundaries between CTUs in the image. The processing order of the CTUs in a slice (an encoding order or a decoding order) is, for example, a raster-scan order. A slice includes header information and encoded data. Features of the slice may be described in header information. The features include a CTU address of a top CTU in the slice, a slice type, etc.

[0144] A tile is a unit of a rectangular region included in a picture. Each of tiles may be assigned with a number referred to as TileId in raster-scan order.

[0145] FIG. 4B is a conceptual diagram indicating an example of a tile configuration. For example, a picture includes 11×8 CTUs and is split into four tiles of rectangular regions (tiles 1 to 4). When tiles are used, the processing order of CTUs are changed from the processing order in the case where no tile is used. When no tile is used, CTUs in a picture are processed in raster-scan order. When tiles are used, at least one CTU in each of the tiles is processed in raster-scan order. For example, as illustrated in FIG. 4B, the processing order of the CTUs included in tile 1 is the order which starts from the left-end of the first row of tile 1 toward the right-end of the first row of tile 1 and then starts from the left-end of the second row of tile 1 toward the right-end of the second row of tile 1.

[0146] It is to be noted that the one tile may include one or more slices, and one slice may include one or more tiles.

[Subtractor]

[0147] Subtractor **104** subtracts a prediction signal (prediction sample that is input from prediction controller **128** indicated below) from an original signal (original sample) in units of a block input from splitter **102** and split by splitter **102**. In other words, subtractor **104** calculates prediction errors (also referred to as residuals) of a block to be encoded (hereinafter also referred to as a current block). Subtractor **104** then outputs the calculated prediction errors (residuals) to transformer **106**.

[0148] The original signal is a signal which has been input into encoder **100** and represents an image of each picture included in a video (for example, a luma signal and two chroma signals). Hereinafter, a signal representing an image is also referred to as a sample.

[Transformer]

[0149] Transformer **106** transforms prediction errors in spatial domain into transform coefficients

in frequency domain, and outputs the transform coefficients to quantizer **108**. More specifically, transformer **106** applies, for example, a defined discrete cosine transform (DCT) or discrete sine transform (DST) to prediction errors in spatial domain. The defined DCT or DST may be predefined.

[0150] It is to be noted that transformer **106** may adaptively select a transform type from among a plurality of transform types, and transform prediction errors into transform coefficients by using a transform basis function corresponding to the selected transform type. This sort of transform is also referred to as explicit multiple core transform (EMT) or adaptive multiple transform (AMT).

[0151] The transform types include, for example, DCT-II, DCT-V, DCT-VIII, DST-I, and DST-VII. FIG. 5A is a chart indicating transform basis functions for the example transform types. In FIG. 5A, N indicates the number of input pixels. For example, selection of a transform type from among the plurality of transform types may depend on a prediction type (one of intra prediction and inter prediction), and may depend on an intra prediction mode.

[0152] Information indicating whether to apply such EMT or AMT (referred to as, for example, an EMT flag or an AMT flag) and information indicating the selected transform type is normally signaled at the CU level. It is to be noted that the signaling of such information does not necessarily need to be performed at the CU level, and may be performed at another level (for example, at the bit sequence level, picture level, slice level, tile level, or CTU level).

[0153] In addition, transformer **106** may re-transform the transform coefficients (transform result). Such re-transform is also referred to as adaptive secondary transform (AST) or non-separable secondary transform (NSST). For example, transformer **106** performs re-transform in units of a sub-block (for example, 4×4 sub-block) included in a transform coefficient block corresponding to an intra prediction error. Information indicating whether to apply NSST and information related to a transform matrix for use in NSST are normally signaled at the CU level. It is to be noted that the signaling of such information does not necessarily need to be performed at the CU level, and may be performed at another level (for example, at the sequence level, picture level, slice level, tile level, or CTU level).

[0154] Transformer **106** may employ a separable transform and a non-separable transform. A separable transform is a method in which a transform is performed a plurality of times by separately performing a transform for each of a number of directions according to the number of dimensions of inputs. A non-separable transform is a method of performing a collective transform in which two or more dimensions in multidimensional inputs are collectively regarded as a single dimension.

[0155] In one example of a non-separable transform, when an input is a 4×4 block, the 4×4 block is regarded as a single array including sixteen elements, and the transform applies a 16×16 transform matrix to the array.

[0156] In another example of a non-separable transform, a 4×4 input block is regarded as a single array including sixteen elements, and then a transform (hypercube givens transform) in which givens revolution is performed on the array a plurality of times may be performed.

[0157] In the transform in transformer **106**, the types of bases to be transformed into the frequency domain according to regions in a CU can be switched. Examples include spatially varying transforms (SVT). In SVT, as illustrated in FIG. 5B, CUs are split into two equal regions horizontally or vertically, and only one of the regions is transformed into the frequency domain. A transform basis type can be set for each region. For example, DST7 and DST8 are used. In this example, only one of these two regions in the CU is transformed, and the other is not transformed. However, both of these two regions may be transformed. In addition, the splitting method is not limited to the splitting into two equal regions, and can be more flexible. For example, the CU may be split into four equal regions, or information indicating splitting may be encoded separately and be signaled in the same manner as the CU splitting. It is to be noted that SVT is also referred to as sub-block transform (SBT).

[Quantizer]

[0158] Quantizer **108** quantizes the transform coefficients output from transformer **106**. More specifically, quantizer **108** scans, in a determined scanning order, the transform coefficients of the current block, and quantizes the scanned transform coefficients based on quantization parameters (QP) corresponding to the transform coefficients. Quantizer **108** then outputs the quantized transform coefficients (hereinafter also referred to as quantized coefficients) of the current block to entropy encoder **110** and inverse quantizer **112**. The determined scanning order may be predetermined.

[0159] A determined scanning order is an order for quantizing/inverse quantizing transform coefficients. For example, a determined scanning order may be defined as ascending order of frequency (from low to high frequency) or descending order of frequency (from high to low frequency).

[0160] A quantization parameter (QP) is a parameter defining a quantization step (quantization width). For example, when the value of the quantization parameter increases, the quantization step also increases. In other words, when the value of the quantization parameter increases, the quantization error increases.

[0161] In addition, a quantization matrix may be used for quantization. For example, several kinds of quantization matrices may be used correspondingly to frequency transform sizes such as 4×4 and 8×8 , prediction modes such as intra prediction and inter prediction, and pixel components such as luma and chroma pixel components. It is to be noted that quantization means digitalizing values sampled at determined intervals correspondingly to determined levels. In this technical field, quantization may be referred to using other expressions, such as rounding and scaling, and may employ rounding and scaling. The determined intervals and levels may be predetermined.

[0162] Methods using quantization matrices include a method using a quantization matrix which has been set directly at the encoder side and a method using a quantization matrix which has been set as a default (default matrix). At the encoder side, a quantization matrix suitable for features of an image can be set by directly setting a quantization matrix. This case, however, has a disadvantage of increasing a coding amount for encoding the quantization matrix.

[0163] There is a method for quantizing a high-frequency coefficient and a low-frequency coefficient without using a quantization matrix. It is to be noted that this method is equivalent to a method using a quantization matrix (flat matrix) whose coefficients have the same value.

[0164] The quantization matrix may be specified using, for example, a sequence parameter set (SPS) or a picture parameter set (PPS). The SPS includes a parameter which is used for a sequence, and the PPS includes a parameter which is used for a picture. Each of the SPS and the PPS may be simply referred to as a parameter set.

[Entropy Encoder]

[0165] Entropy encoder **110** generates an encoded signal (encoded bitstream) based on quantized coefficients which have been input from quantizer **108**. More specifically, entropy encoder **110**, for example, binarizes quantized coefficients, and arithmetically encodes the binary signal, and outputs a compressed bit stream or sequence.

[Inverse Quantizer]

[0166] Inverse quantizer **112** inverse quantizes quantized coefficients which have been input from quantizer **108**. More specifically, inverse quantizer **112** inverse quantizes, in a determined scanning order, quantized coefficients of the current block. Inverse quantizer **112** then outputs the inverse quantized transform coefficients of the current block to inverse transformer **114**. The determined scanning order may be predetermined.

[Inverse Transformer]

[0167] Inverse transformer **114** restores prediction errors (residuals) by inverse transforming transform coefficients which have been input from inverse quantizer **112**. More specifically, inverse transformer **114** restores the prediction errors of the current block by applying an inverse transform

corresponding to the transform applied by transformer **106** on the transform coefficients. Inverse transformer **114** then outputs the restored prediction errors to adder **116**.

[0168] It is to be noted that since information is lost in quantization, the restored prediction errors do not match the prediction errors calculated by subtractor **104**. In other words, the restored prediction errors normally include quantization errors.

[Adder]

[0169] Adder **116** reconstructs the current block by adding prediction errors which have been input from inverse transformer **114** and prediction samples which have been input from prediction controller **128**. Adder **116** then outputs the reconstructed block to block memory **118** and loop filter **120**. A reconstructed block is also referred to as a local decoded block.

[Block Memory]

[0170] Block memory **118** is, for example, storage for storing blocks in a picture to be encoded (hereinafter referred to as a current picture) which is referred to in intra prediction. More specifically, block memory **118** stores reconstructed blocks output from adder **116**.

[Frame Memory]

[0171] Frame memory **122** is, for example, storage for storing reference pictures for use in inter prediction, and is also referred to as a frame buffer. More specifically, frame memory **122** stores reconstructed blocks filtered by loop filter **120**.

[Loop Filter]

[0172] Loop filter **120** applies a loop filter to blocks reconstructed by adder **116**, and outputs the filtered reconstructed blocks to frame memory **122**. A loop filter is a filter used in an encoding loop (in-loop filter), and includes, for example, a deblocking filter (DF or DBF), a sample adaptive offset (SAO), and an adaptive loop filter (ALF).

[0173] In an ALF, a least square error filter for removing compression artifacts is applied. For example, one filter selected from among a plurality of filters based on the direction and activity of local gradients is applied for each of 2×2 sub-blocks in the current block.

[0174] More specifically, first, each sub-block (for example, each 2×2 sub-block) is categorized into one out of a plurality of classes (for example, fifteen or twenty-five classes). The classification of the sub-block is based on gradient directionality and activity. For example, classification index C (for example, $C=5D+A$) is derived based on gradient directionality D (for example, 0 to 2 or 0 to 4) and gradient activity A (for example, 0 to 4). Then, based on classification index C, each sub-block is categorized into one out of a plurality of classes.

[0175] For example, gradient directionality D is calculated by comparing gradients of a plurality of directions (for example, the horizontal, vertical, and two diagonal directions). Moreover, for example, gradient activity A is calculated by adding gradients of a plurality of directions and quantizing the result of addition.

[0176] The filter to be used for each sub-block is determined from among the plurality of filters based on the result of such categorization.

[0177] The filter shape to be used in an ALF is, for example, a circular symmetric filter shape. FIG. 6A through FIG. 6C illustrate examples of filter shapes used in ALFs. FIG. 6A illustrates a 5×5 diamond shape filter, FIG. 6B illustrates a 7×7 diamond shape filter, and FIG. 6C illustrates a 9×9 diamond shape filter. Information indicating the filter shape is normally signaled at the picture level. It is to be noted that the signaling of such information indicating the filter shape does not necessarily need to be performed at the picture level, and may be performed at another level (for example, at the sequence level, slice level, tile level, CTU level, or CU level).

[0178] The ON or OFF of the ALF is determined, for example, at the picture level or CU level. For example, the decision of whether to apply the ALF to luma may be made at the CU level, and the decision of whether to apply ALF to chroma may be made at the picture level. Information indicating ON or OFF of the ALF is normally signaled at the picture level or CU level. It is to be noted that the signaling of information indicating ON or OFF of the ALF does not necessarily need

to be performed at the picture level or CU level, and may be performed at another level (for example, at the sequence level, slice level, tile level, or CTU level).

[0179] The coefficient set for the plurality of selectable filters (for example, fifteen or up to twenty-five filters) is normally signaled at the picture level. It is to be noted that the signaling of the coefficient set does not necessarily need to be performed at the picture level, and may be performed at another level (for example, at the sequence level, slice level, tile level, CTU level, CU level, or sub-block level).

[Loop Filter>Deblocking Filter]

[0180] In a deblocking filter, loop filter **120** performs a filter process on a block boundary in a reconstructed image so as to reduce distortion which occurs at the block boundary.

[0181] FIG. 7 is a block diagram illustrating one example of a specific configuration of loop filter **120** which functions as a deblocking filter.

[0182] Loop filter **120** includes: boundary determiner **1201**; filter determiner **1203**; filtering executor **1205**; process determiner **1208**; filter characteristic determiner **1207**; and switches **1202**, **1204**, and **1206**.

[0183] Boundary determiner **1201** determines whether a pixel to be deblock-filtered (that is, a current pixel) is present around a block boundary. Boundary determiner **1201** then outputs the determination result to switch **1202** and processing determiner **1208**.

[0184] In the case where boundary determiner **1201** has determined that a current pixel is present around a block boundary, switch **1202** outputs an unfiltered image to switch **1204**. In the opposite case where boundary determiner **1201** has determined that no current pixel is present around a block boundary, switch **1202** outputs an unfiltered image to switch **1206**.

[0185] Filter determiner **1203** determines whether to perform deblocking filtering of the current pixel, based on the pixel value of at least one surrounding pixel located around the current pixel. Filter determiner **1203** then outputs the determination result to switch **1204** and processing determiner **1208**.

[0186] In the case where filter determiner **1203** has determined to perform deblocking filtering of the current pixel, switch **1204** outputs the unfiltered image obtained through switch **1202** to filtering executor **1205**. In the opposite case where filter determiner **1203** has determined not to perform deblocking filtering of the current pixel, switch **1204** outputs the unfiltered image obtained through switch **1202** to switch **1206**.

[0187] When obtaining the unfiltered image through switches **1202** and **1204**, filtering executor **1205** executes, for the current pixel, deblocking filtering with the filter characteristic determined by filter characteristic determiner **1207**. Filtering executor **1205** then outputs the filtered pixel to switch **1206**.

[0188] Under control by processing determiner **1208**, switch **1206** selectively outputs a pixel which has not been deblock-filtered and a pixel which has been deblock-filtered by filtering executor **1205**.

[0189] Processing determiner **1208** controls switch **1206** based on the results of determinations made by boundary determiner **1201** and filter determiner **1203**. In other words, processing determiner **1208** causes switch **1206** to output the pixel which has been deblock-filtered when boundary determiner **1201** has determined that the current pixel is present around the block boundary and filter determiner **1203** has determined to perform deblocking filtering of the current pixel. In addition, other than the above case, processing determiner **1208** causes switch **1206** to output the pixel which has not been deblock-filtered. A filtered image is output from switch **1206** by repeating output of a pixel in this way.

[0190] FIG. 8 is a conceptual diagram indicating an example of a deblocking filter having a symmetrical filtering characteristic with respect to a block boundary.

[0191] In a deblocking filter process, one of two deblocking filters having different characteristics, that is, a strong filter and a weak filter is selected using pixel values and quantization parameters.

In the case of the strong filter, pixels p0 to p2 and pixels q0 to q2 are present across a block boundary as illustrated in FIG. 8, the pixel values of the respective pixel q0 to q2 are changed to pixel values q'0 to q'2 by performing, for example, computations according to the expressions below.

$$[00001] q'0 = (p1 + 2 \times p0 + 2 \times q0 + 2 \times q1 + q2 + 4) / 8$$
$$q'1 = (p0 + q0 + q1 + q2 + 2) / 4$$
$$q'2 = (p0 + q0 + q1 + 3 \times q2 + 2 \times q3 + 4) / 8$$

[0192] It is to be noted that, in the above expressions, p0 to p2 and q0 to q2 are the pixel values of respective pixels p0 to p2 and pixels q0 to q2. In addition, q3 is the pixel value of neighboring pixel q3 located at the opposite side of pixel q2 with respect to the block boundary. In addition, in the right side of each of the expressions, coefficients which are multiplied with the respective pixel values of the pixels to be used for deblocking filtering are filter coefficients.

[0193] Furthermore, in the deblocking filtering, clipping may be performed so that the calculated pixel values are not set over a threshold value. In the clipping process, the pixel values calculated according to the above expressions are clipped to a value obtained according to “a computation pixel value $\pm 2 \times$ a threshold value” using the threshold value determined based on a quantization parameter. In this way, it is possible to prevent excessive smoothing.

[0194] FIG. 9 is a conceptual diagram for illustrating a block boundary on which a deblocking filter process is performed. FIG. 10 is a conceptual diagram indicating examples of Bs values.

[0195] The block boundary on which the deblocking filter process is performed is, for example, a boundary between prediction units (PU) having 8×8 pixel blocks as illustrated in FIG. 9 or a boundary between transform units (TU). The deblocking filter process may be performed in units of four rows or four columns. First, boundary strength (Bs) values are determined as indicated in FIG. 10 for block P and block Q illustrated in FIG. 9.

[0196] According to the Bs values in FIG. 10, whether to perform deblocking filter processes of block boundaries belonging to the same image using different strengths is determined. The deblocking filter process for a chroma signal is performed when a Bs value is 2. The deblocking filter process for a luma signal is performed when a Bs value is 1 or more and a determined condition is satisfied. The determined condition may be predetermined. It is to be noted that conditions for determining Bs values are not limited to those indicated in FIG. 10, and a Bs value may be determined based on another parameter.

[Prediction Processor (Intra Predictor, Inter Predictor, Prediction Controller)]

[0197] FIG. 11 is a flow chart illustrating one example of a process performed by the prediction processor of encoder 100. It is to be noted that the prediction processor includes all or part of the following constituent elements: intra predictor 124; inter predictor 126; and prediction controller 128.

[0198] The prediction processor generates a prediction image of a current block (Step Sb_1). This prediction image is also referred to as a prediction signal or a prediction block. It is to be noted that the prediction signal is, for example, an intra prediction signal or an inter prediction signal. Specifically, the prediction processor generates the prediction image of the current block using a reconstructed image which has been already obtained through generation of a prediction block, generation of a difference block, generation of a coefficient block, restoring of a difference block, and generation of a decoded image block.

[0199] The reconstructed image may be, for example, an image in a reference picture, or an image of an encoded block in a current picture which is the picture including the current block. The encoded block in the current picture is, for example, a neighboring block of the current block.

[0200] FIG. 12 is a flow chart illustrating another example of a process performed by the prediction processor of encoder 100.

[0201] The prediction processor generates a prediction image using a first method (Step Sc_1a), generates a prediction image using a second method (Step Sc_1b), and generates a prediction image using a third method (Step Sc_1c). The first method, the second method, and the third

method may be mutually different methods for generating a prediction image. Each of the first to third methods may be an inter prediction method, an intra prediction method, or another prediction method. The above-described reconstructed image may be used in these prediction methods.

[0202] Next, the prediction processor selects any one of a plurality of prediction methods generated in Steps Sc_1a, Sc_1b, and Sc_1c (Step Sc_2). The selection of the prediction image, that is selection of a method or a mode for obtaining a final prediction image may be made by calculating a cost for each of the generated prediction images and based on the cost. Alternatively, the selection of the prediction image may be made based on a parameter which is used in an encoding process. Encoder **100** may transform information for identifying a selected prediction image, a method, or a mode into an encoded signal (also referred to as an encoded bitstream). The information may be, for example, a flag or the like. In this way, the decoder is capable of generating a prediction image according to the method or the mode selected based on the information in encoder **100**. It is to be noted that, in the example illustrated in FIG. **12**, the prediction processor selects any of the prediction images after the prediction images are generated using the respective methods. However, the prediction processor may select a method or a mode based on a parameter for use in the above-described encoding process before generating prediction images, and may generate a prediction image according to the method or mode selected.

[0203] For example, the first method and the second method may be intra prediction and inter prediction, respectively, and the prediction processor may select a final prediction image for a current block from prediction images generated according to the prediction methods.

[0204] FIG. **13** is a flow chart illustrating another example of a process performed by the prediction processor of encoder **100**.

[0205] First, the prediction processor generates a prediction image using intra prediction (Step Sd_1a), and generates a prediction image using inter prediction (Step Sd_1b). It is to be noted that the prediction image generated by intra prediction is also referred to as an intra prediction image, and the prediction image generated by inter prediction is also referred to as an inter prediction image.

[0206] Next, the prediction processor evaluates each of the intra prediction image and the inter prediction image (Step Sd_2). A cost may be used in the evaluation. In other words, the prediction processor calculates cost C for each of the intra prediction image and the inter prediction image. Cost C may be calculated according to an expression of an R-D optimization model, for example, $C=D+\lambda \times R$. In this expression, D indicates a coding distortion of a prediction image, and is represented as, for example, a sum of absolute differences between the pixel value of a current block and the pixel value of a prediction image. In addition, R indicates a predicted coding amount of a prediction image, specifically, the coding amount required to encode motion information for generating a prediction image, etc. In addition, λ indicates, for example, a multiplier according to the method of Lagrange multiplier.

[0207] The prediction processor then selects the prediction image for which the smallest cost C has been calculated among the intra prediction image and the inter prediction image, as the final prediction image for the current block (Step Sd_3). In other words, the prediction method or the mode for generating the prediction image for the current block is selected.

[Intra Predictor]

[0208] Intra predictor **124** generates a prediction signal (intra prediction signal) by performing intra prediction (also referred to as intra frame prediction) of the current block by referring to a block or blocks in the current picture and stored in block memory **118**. More specifically, intra predictor **124** generates an intra prediction signal by performing intra prediction by referring to samples (for example, luma and/or chroma values) of a block or blocks neighboring the current block, and then outputs the intra prediction signal to prediction controller **128**.

[0209] For example, intra predictor **124** performs intra prediction by using one mode from among a plurality of intra prediction modes which have been defined. The intra prediction modes include

one or more non-directional prediction modes and a plurality of directional prediction modes. The defined modes may be predefined.

[0210] The one or more non-directional prediction modes include, for example, the planar prediction mode and DC prediction mode defined in the H.265/high-efficiency video coding (HEVC) standard.

[0211] The plurality of directional prediction modes include, for example, the thirty-three directional prediction modes defined in the H.265/HEVC standard. It is to be noted that the plurality of directional prediction modes may further include thirty-two directional prediction modes in addition to the thirty-three directional prediction modes (for a total of sixty-five directional prediction modes). FIG. 14 is a conceptual diagram illustrating sixty-seven intra prediction modes in total that may be used in intra prediction (two non-directional prediction modes and sixty-five directional prediction modes). The solid arrows represent the thirty-three directions defined in the H.265/HEVC standard, and the dashed arrows represent the additional thirty-two directions (the two non-directional prediction modes are not illustrated in FIG. 14).

[0212] In various kinds of processing examples, a luma block may be referred to in intra prediction of a chroma block. In other words, a chroma component of the current block may be predicted based on a luma component of the current block. Such intra prediction is also referred to as cross-component linear model (CCLM) prediction. The intra prediction mode for a chroma block in which such a luma block is referred to (also referred to as, for example, a CCLM mode) may be added as one of the intra prediction modes for chroma blocks.

[0213] Intra predictor **124** may correct intra-predicted pixel values based on horizontal/vertical reference pixel gradients. Intra prediction accompanied by this sort of correcting is also referred to as position dependent intra prediction combination (PDPC). Information indicating whether to apply PDPC (referred to as, for example, a PDPC flag) is normally signaled at the CU level. It is to be noted that the signaling of such information does not necessarily need to be performed at the CU level, and may be performed at another level (for example, at the sequence level, picture level, slice level, tile level, or CTU level).

[Inter Predictor]

[0214] Inter predictor **126** generates a prediction signal (inter prediction signal) by performing inter prediction (also referred to as inter frame prediction) of the current block by referring to a block or blocks in a reference picture, which is different from the current picture and is stored in frame memory **122**. Inter prediction is performed in units of a current block or a current sub-block (for example, a 4×4 block) in the current block. For example, inter predictor **126** performs motion estimation in a reference picture for the current block or the current sub-block, and finds out a reference block or a sub-block which best matches the current block or the current sub-block. Inter predictor **126** then obtains motion information (for example, a motion vector) which compensates a motion or a change from the reference block or the sub-block to the current block or the sub-block. Inter predictor **126** generates an inter prediction signal of the current block or the sub-block by performing motion compensation (or motion prediction) based on the motion information. Inter predictor **126** outputs the generated inter prediction signal to prediction controller **128**.

[0215] The motion information used in motion compensation may be signaled as inter prediction signals in various forms. For example, a motion vector may be signaled. As another example, the difference between a motion vector and a motion vector predictor may be signaled.

[Basic Flow of Inter Prediction]

[0216] FIG. 15 is a flow chart illustrating an example basic processing flow of inter prediction.

[0217] First, inter predictor **126** generates a prediction signal (Steps Se_1 to Se_3). Next, subtractor **104** generates the difference between a current block and a prediction image as a prediction residual (Step Se_4).

[0218] Here, in the generation of the prediction image, inter predictor **126** generates the prediction image through determination of a motion vector (MV) of the current block (Steps Se_1 and Se_2)

and motion compensation (Step Se_3). Furthermore, in determination of an MV, inter predictor **126** determines the MV through selection of a motion vector candidate (MV candidate) (Step Se_1) and derivation of an MV (Step Se_2). The selection of the MV candidate is made by, for example, selecting at least one MV candidate from an MV candidate list. Alternatively, in derivation of an MV, inter predictor **126** may further select at least one MV candidate from the at least one MV candidate, and determine the selected at least one MV candidate as the MV for the current block. Alternatively, inter predictor **126** may determine the MV for the current block by performing estimation in a reference picture region specified by each of the selected at least one MV candidate. It is to be noted that the estimation in a reference picture region may be referred to as motion estimation.

[0219] In addition, although Steps Se_1 to Se_3 are performed by inter predictor **126** in the above-described example, a process that is for example Step Se_1, Step Se_2, or the like may be performed by another constituent element included in encoder **100**.

[Motion Vector Derivation Flow]

[0220] FIG. **16** is a flow chart illustrating one example of derivation of motion vectors.

[0221] Inter predictor **126** derives an MV of a current block in a mode for encoding motion information (for example, an MV). In this case, for example, the motion information is encoded as a prediction parameter, and is signaled. In other words, the encoded motion information is included in an encoded signal (also referred to as an encoded bitstream).

[0222] Alternatively, inter predictor **126** derives an MV in a mode in which motion information is not encoded. In this case, no motion information is included in an encoded signal.

[0223] Here, MV derivation modes may include a normal inter mode, a merge mode, a FRUC mode, an affine mode, etc. which are described later. Modes in which motion information is encoded among the modes include the normal inter mode, the merge mode, the affine mode (specifically, an affine inter mode and an affine merge mode), etc. It is to be noted that motion information may include not only an MV but also motion vector predictor selection information which is described later. Modes in which no motion information is encoded include the FRUC mode, etc. Inter predictor **126** selects a mode for deriving an MV of the current block from the modes, and derives the MV of the current block using the selected mode.

[0224] FIG. **17** is a flow chart illustrating another example of derivation of motion vectors.

[0225] Inter predictor **126** derives an MV of a current block in a mode in which an MV difference is encoded. In this case, for example, the MV difference is encoded as a prediction parameter, and is signaled. In other words, the encoded MV difference is included in an encoded signal. The MV difference is the difference between the MV of the current block and the MV predictor.

[0226] Alternatively, inter predictor **126** derives an MV in a mode in which no MV difference is encoded. In this case, no encoded MV difference is included in an encoded signal.

[0227] Here, as described above, the MV derivation modes include the normal inter mode, the merge mode, the FRUC mode, the affine mode, etc. which are described later. Modes in which an MV difference is encoded among the modes include the normal inter mode, the affine mode (specifically, the affine inter mode), etc. Modes in which no MV difference is encoded include the FRUC mode, the merge mode, the affine mode (specifically, the affine merge mode), etc. Inter predictor **126** selects a mode for deriving an MV of the current block from the plurality of modes, and derives the MV of the current block using the selected mode.

[Motion Vector Derivation Flow]

[0228] FIG. **18** is a flow chart illustrating another example of derivation of motion vectors. The MV derivation modes which are inter prediction modes include a plurality of modes and are roughly divided into modes in which an MV difference is encoded and modes in which no motion vector difference is encoded. The modes in which no MV difference is encoded include the merge mode, the FRUC mode, the affine mode (specifically, the affine merge mode), etc. These modes are described in detail later. Simply, the merge mode is a mode for deriving an MV of a current block

by selecting a motion vector from an encoded surrounding block, and the FRUC mode is a mode for deriving an MV of a current block by performing estimation between encoded regions. The affine mode is a mode for deriving, as an MV of a current block, a motion vector of each of a plurality of sub-blocks included in the current block, assuming affine transform.

[0229] More specifically, as illustrated when the inter prediction mode information indicates 0 (0 in Sf_1), inter predictor **126** derives a motion vector using the merge mode (Sf_2). When the inter prediction mode information indicates 1 (1 in Sf_1), inter predictor **126** derives a motion vector using the FRUC mode (Sf_3). When the inter prediction mode information indicates 2 (2 in Sf_1), inter predictor **126** derives a motion vector using the affine mode (specifically, the affine merge mode) (Sf_4). When the inter prediction mode information indicates 3 (3 in Sf_1), inter predictor **126** derives a motion vector using a mode in which an MV difference is encoded (for example, a normal inter mode (Sf_5)).

[MV Derivation>Normal Inter Mode]

[0230] The normal inter mode is an inter prediction mode for deriving an MV of a current block based on a block similar to the image of the current block from a reference picture region specified by an MV candidate. In this normal inter mode, an MV difference is encoded.

[0231] FIG. **19** is a flow chart illustrating an example of inter prediction in normal inter mode.

[0232] First, inter predictor **126** obtains a plurality of MV candidates for a current block based on information such as MVs of a plurality of encoded blocks temporally or spatially surrounding the current block (Step Sg_1). In other words, inter predictor **126** generates an MV candidate list.

[0233] Next, inter predictor **126** extracts N (an integer of 2 or larger) MV candidates from the plurality of MV candidates obtained in Step Sg_1, as motion vector predictor candidates (also referred to as MV predictor candidates) according to a determined priority order (Step Sg_2). It is to be noted that the priority order may be determined in advance for each of the N MV candidates.

[0234] Next, inter predictor **126** selects one motion vector predictor candidate from the N motion vector predictor candidates, as the motion vector predictor (also referred to as an MV predictor) of the current block (Step Sg_3). At this time, inter predictor **126** encodes, in a stream, motion vector predictor selection information for identifying the selected motion vector predictor. It is to be noted that the stream is an encoded signal or an encoded bitstream as described above.

[0235] Next, inter predictor **126** derives an MV of a current block by referring to an encoded reference picture (Step Sg_4). At this time, inter predictor **126** further encodes, in the stream, the difference value between the derived MV and the motion vector predictor as an MV difference. It is to be noted that the encoded reference picture is a picture including a plurality of blocks which have been reconstructed after being encoded.

[0236] Lastly, inter predictor **126** generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the encoded reference picture (Step Sg_5). It is to be noted that the prediction image is an inter prediction signal as described above.

[0237] In addition, information indicating the inter prediction mode (normal inter mode in the above example) used to generate the prediction image is, for example, encoded as a prediction parameter.

[0238] It is to be noted that the MV candidate list may be also used as a list for use in another mode. In addition, the processes related to the MV candidate list may be applied to processes related to the list for use in another mode. The processes related to the MV candidate list include, for example, extraction or selection of an MV candidate from the MV candidate list, reordering of MV candidates, or deletion of an MV candidate.

[MV Derivation>Merge Mode]

[0239] The merge mode is an inter prediction mode for selecting an MV candidate from an MV candidate list as an MV of a current block, thereby deriving the MV.

[0240] FIG. **20** is a flow chart illustrating an example of inter prediction in merge mode.

[0241] First, inter predictor **126** obtains a plurality of MV candidates for a current block based on information such as MVs of a plurality of encoded blocks temporally or spatially surrounding the current block (Step Sh_1). In other words, inter predictor **126** generates an MV candidate list.

[0242] Next, inter predictor **126** selects one MV candidate from the plurality of MV candidates obtained in Step Sh_1, thereby deriving an MV of the current block (Step Sh_2). At this time, inter predictor **126** encodes, in a stream, MV selection information for identifying the selected MV candidate.

[0243] Lastly, inter predictor **126** generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the encoded reference picture (Step Sh_3).

[0244] In addition, information indicating the inter prediction mode (merge mode in the above example) used to generate the prediction image and included in the encoded signal is, for example, encoded as a prediction parameter.

[0245] FIG. **21** is a conceptual diagram for illustrating one example of a motion vector derivation process of a current picture in merge mode.

[0246] First, an MV candidate list in which MV predictor candidates are registered is generated. Examples of MV predictor candidates include: spatially neighboring MV predictors which are MVs of a plurality of encoded blocks located spatially surrounding a current block; temporally neighboring MV predictors which are MVs of surrounding blocks on which the position of a current block in an encoded reference picture is projected; combined MV predictors which are MVs generated by combining the MV value of a spatially neighboring MV predictor and the MV of a temporally neighboring MV predictor; and a zero MV predictor which is an MV having a zero value.

[0247] Next, one MV predictor is selected from a plurality of MV predictors registered in an MV predictor list, and the selected MV predictor is determined as the MV of a current block.

[0248] Furthermore, the variable length encoder describes and encodes, in a stream, merge_idx which is a signal indicating which MV predictor has been selected.

[0249] It is to be noted that the MV predictors registered in the MV predictor list described in FIG. **21** are examples. The number of MV predictors may be different from the number of MV predictors in the diagram, the MV predictor list may be configured in such a manner that some of the kinds of the MV predictors in the diagram may not be included, or that one or more MV predictors other than the kinds of MV predictors in the diagram are included.

[0250] A final MV may be determined by performing a decoder motion vector refinement process (DMVR) to be described later using the MV of the current block derived in merge mode.

[0251] It is to be noted that the MV predictor candidates are MV candidates described above, and the MV predictor list is the MV candidate list described above. It is to be noted that the MV candidate list may be referred to as a candidate list. In addition, merge_idx is MV selection information.

[MV Derivation>FRUC Mode]

[0252] Motion information may be derived at the decoder side without being signaled from the encoder side. It is to be noted that, as described above, the merge mode defined in the H.265/HEVC standard may be used. In addition, for example, motion information may be derived by performing motion estimation at the decoder side. In an embodiment, at the decoder side, motion estimation is performed without using any pixel value in a current block.

[0253] Here, a mode for performing motion estimation at the decoder side is described. The mode for performing motion estimation at the decoder side may be referred to as a pattern matched motion vector derivation (PMMVD) mode, or a frame rate up-conversion (FRUC) mode.

[0254] One example of a FRUC process in the form of a flow chart is illustrated in FIG. **22**. First, a list of a plurality of candidates each having a motion vector (MV) predictor (that is, an MV candidate list that may be also used as a merge list) is generated by referring to a motion vector in

an encoded block which is spatially or temporally neighbors a current block (Step Si_1). Next, a best MV candidate is selected from the plurality of MV candidates registered in the MV candidate list (Step Si_2). For example, the evaluation values of the respective MV candidates included in the MV candidate list are calculated, and one MV candidate is selected based on the evaluation values. Based on the selected motion vector candidates, a motion vector for the current block is then derived (Step Si_4). More specifically, for example, the selected motion vector candidate (best MV candidate) is derived directly as the motion vector for the current block. In addition, for example, the motion vector for the current block may be derived using pattern matching in a surrounding region of a position in a reference picture where the position in the reference picture corresponds to the selected motion vector candidate. In other words, estimation using the pattern matching and the evaluation values may be performed in the surrounding region of the best MV candidate, and when there is an MV that yields a better evaluation value, the best MV candidate may be updated to the MV that yields the better evaluation value, and the updated MV may be determined as the final MV for the current block. A configuration in which no such a process for updating the best MV candidate to the MV having a better evaluation value is performed is also possible.

[0255] Lastly, inter predictor **126** generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the encoded reference picture (Step Si_5).

[0256] A similar process may be performed in units of a sub-block.

[0257] Evaluation values may be calculated according to various kinds of methods. For example, a comparison is made between a reconstructed image in a region in a reference picture corresponding to a motion vector and a reconstructed image in a determined region (the region may be, for example, a region in another reference picture or a region in a neighboring block of a current picture, as indicated below). The determined region may be predetermined.

[0258] The difference between the pixel values of the two reconstructed images may be used for an evaluation value of the motion vectors. It is to be noted that an evaluation value may be calculated using information other than the value of the difference.

[0259] Next, an example of pattern matching is described in detail. First, one MV candidate included in an MV candidate list (for example, a merge list) is selected as a start point of estimation by the pattern matching. For example, as the pattern matching, either a first pattern matching or a second pattern matching may be used. The first pattern matching and the second pattern matching are also referred to as bilateral matching and template matching, respectively.

[MV Derivation>FRUC>Bilateral Matching]

[0260] In the first pattern matching, pattern matching is performed between two blocks along a motion trajectory of a current block which are two blocks in different two reference pictures. Accordingly, in the first pattern matching, a region in another reference picture along the motion trajectory of the current block is used as a determined region for calculating the evaluation value of the above-described candidate. The determined region may be predetermined.

[0261] FIG. 23 is a conceptual diagram for illustrating one example of the first pattern matching (bilateral matching) between the two blocks in the two reference pictures along the motion trajectory. As illustrated in FIG. 23, in the first pattern matching, two motion vectors (MV0, MV1) are derived by estimating a pair which best matches among pairs in the two blocks in the two different reference pictures (Ref0, Ref1) which are the two blocks along the motion trajectory of the current block (Cur block). More specifically, a difference between the reconstructed image at a specified location in the first encoded reference picture (Ref0) specified by an MV candidate and the reconstructed image at a specified location in the second encoded reference picture (Ref1) specified by a symmetrical MV obtained by scaling the MV candidate at a display time interval is derived for the current block, and an evaluation value is calculated using the value of the obtained difference. It is possible to select, as the final MV, the MV candidate which yields the best evaluation value among the plurality of MV candidates, and which is likely to produce good

results.

[0262] In the assumption of a continuous motion trajectory, the motion vectors (MV0, MV1) specifying the two reference blocks are proportional to temporal distances (TD0, TD1) between the current picture (Cur Pic) and the two reference pictures (Ref0, Ref1). For example, when the current picture is temporally located between the two reference pictures and the temporal distances from the current picture to the respective two reference pictures are equal to each other, mirror-symmetrical bi-directional motion vectors are derived in the first pattern matching.

[MV Derivation>FRUC>Template Matching]

[0263] In the second pattern matching (template matching), pattern matching is performed between a block in a reference picture and a template in the current picture (the template is a block neighboring the current block in the current picture (the neighboring block is, for example, an upper and/or left neighboring block(s))). Accordingly, in the second pattern matching, the block neighboring the current block in the current picture is used as the determined region for calculating the evaluation value of the above-described candidate.

[0264] FIG. 24 is a conceptual diagram for illustrating one example of pattern matching (template matching) between a template in a current picture and a block in a reference picture. As illustrated in FIG. 24, in the second pattern matching, the motion vector of the current block (Cur block) is derived by estimating, in the reference picture (Ref0), the block which best matches the block neighboring the current block in the current picture (Cur Pic). More specifically, it is possible that the difference between a reconstructed image in an encoded region which neighbors both left and above or either left or above and a reconstructed image which is in a corresponding region in the encoded reference picture (Ref0) and is specified by an MV candidate is derived, an evaluation value is calculated using the value of the obtained difference, and the MV candidate which yields the best evaluation value among a plurality of MV candidates is selected as the best MV candidate.

[0265] Such information indicating whether to apply the FRUC mode (referred to as, for example, a FRUC flag) may be signaled at the CU level. In addition, when the FRUC mode is applied (for example, when a FRUC flag is true), information indicating an applicable pattern matching method (either the first pattern matching or the second pattern matching) may be signaled at the CU level. It is to be noted that the signaling of such information does not necessarily need to be performed at the CU level, and may be performed at another level (for example, at the sequence level, picture level, slice level, tile level, CTU level, or sub-block level).

[MV Derivation>Affine Mode]

[0266] Next, the affine mode for deriving a motion vector in units of a sub-block based on motion vectors of a plurality of neighboring blocks is described. This mode is also referred to as an affine motion compensation prediction mode.

[0267] FIG. 25A is a conceptual diagram for illustrating one example of deriving a motion vector of each sub-block based on motion vectors of a plurality of neighboring blocks. In FIG. 25A, the current block includes sixteen 4×4 sub-blocks. Here, motion vector V.sub.0 at an upper-left corner control point in the current block is derived based on a motion vector of a neighboring block, and likewise, motion vector V.sub.1 at an upper-right corner control point in the current block is derived based on a motion vector of a neighboring sub-block. Two motion vectors v.sub.0 and v.sub.1 may be projected according to an expression (1A) indicated below, and motion vectors (v.sub.x, v.sub.y) for the respective sub-blocks in the current block may be derived.

[Math . 1]

$$[00002] \quad \begin{cases} v_x = \frac{(v_{1x} - v_{0x})}{w}x - \frac{(v_{1y} - v_{0y})}{w}y + v_{0x} \\ v_y = \frac{(v_{1y} - v_{0y})}{w}x - \frac{(v_{1x} - v_{0x})}{w}y + v_{0y} \end{cases} \quad (1A)$$

[0268] Here, x and y indicate the horizontal position and the vertical position of the sub-block, respectively, and w indicates a determined weighting coefficient. The determined weighting

coefficient may be predetermined.

[0269] Such information indicating the affine mode (for example, referred to as an affine flag) may be signaled at the CU level. It is to be noted that the signaling of the information indicating the affine mode does not necessarily need to be performed at the CU level, and may be performed at another level (for example, at the sequence level, picture level, slice level, tile level, CTU level, or sub-block level).

[0270] In addition, the affine mode may include several modes for different methods for deriving motion vectors at the upper-left and upper-right corner control points. For example, the affine mode include two modes which are the affine inter mode (also referred to as an affine normal inter mode) and the affine merge mode.

[MV Derivation>Affine Mode]

[0271] FIG. 25B is a conceptual diagram for illustrating one example of deriving a motion vector of each sub-block in affine mode in which three control points are used. In FIG. 25B, the current block includes sixteen 4×4 blocks. Here, motion vector V.sub.0 at the upper-left corner control point for the current block is derived based on a motion vector of a neighboring block, and likewise, motion vector V.sub.1 at the upper-right corner control point for the current block is derived based on a motion vector of a neighboring block, and motion vector V.sub.2 at the lower-left corner control point for the current block is derived based on a motion vector of a neighboring block. Three motion vectors v.sub.0, v.sub.1, and v.sub.2 may be projected according to an expression (1B) indicated below, and motion vectors (v.sub.x, v.sub.y) for the respective sub-blocks in the current block may be derived.

[Math . 2]

$$[00003] \quad \begin{cases} v_x = \frac{(v_{1x} - v_{0x})}{w}x - \frac{(v_{2x} - v_{0x})}{h}y + v_{0x} \\ v_y = \frac{(v_{1y} - v_{0y})}{w}x - \frac{(v_{2y} - v_{0y})}{h}y + v_{0y} \end{cases} \quad (1B)$$

[0272] Here, x and y indicate the horizontal position and the vertical position of the center of the sub-block, respectively, w indicates the width of the current block, and h indicates the height of the current block.

[0273] Affine modes in which different numbers of control points (for example, two and three control points) are used may be switched and signaled at the CU level. It is to be noted that information indicating the number of control points in affine mode used at the CU level may be signaled at another level (for example, the sequence level, picture level, slice level, tile level, CTU level, or sub-block level).

[0274] In addition, such an affine mode in which three control points are used may include different methods for deriving motion vectors at the upper-left, upper-right, and lower-left corner control points. For example, the affine modes include two modes which are the affine inter mode (also referred to as the affine normal inter mode) and the affine merge mode.

[MV Derivation>Affine Merge Mode]

[0275] FIG. 26A, FIG. 26B, and FIG. 26C are conceptual diagrams for illustrating the affine merge mode.

[0276] As illustrated in FIG. 26A, in the affine merge mode, for example, motion vector predictors at respective control points of a current block are calculated based on a plurality of motion vectors corresponding to blocks encoded according to the affine mode among encoded block A (left), block B (upper), block C (upper-right), block D (lower-left), and block E (upper-left) which neighbor the current block. More specifically, encoded block A (left), block B (upper), block C (upper-right), block D (lower-left), and block E (upper-left) are checked in the listed order, and the first effective block encoded according to the affine mode is identified. Motion vector predictors at the control points of the current block are calculated based on a plurality of motion vectors corresponding to the identified block.

[0277] For example, as illustrated in FIG. 26B, when block A which neighbors to the left of the current block has been encoded according to an affine mode in which two control points are used, motion vectors $v_{\text{sub.3}}$ and $v_{\text{sub.4}}$ projected at the upper-left corner position and the upper-right corner position of the encoded block including block A are derived. Motion vector predictor $v_{\text{sub.0}}$ at the upper-left corner control point of the current block and motion vector predictor $v_{\text{sub.1}}$ at the upper-right corner control point of the current block are then calculated from derived motion vectors $v_{\text{sub.3}}$ and $v_{\text{sub.4}}$.

[0278] For example, as illustrated in FIG. 26C, when block A which neighbors to the left of the current block has been encoded according to an affine mode in which three control points are used, motion vectors $v_{\text{sub.3}}$, $v_{\text{sub.4}}$, and $v_{\text{sub.5}}$ projected at the upper-left corner position, the upper-right corner position, and the lower-left corner position of the encoded block including block A are derived. Motion vector predictor $v_{\text{sub.0}}$ at the upper-left corner control point of the current block, motion vector predictor $v_{\text{sub.1}}$ at the upper-right corner control point of the current block, and motion vector predictor $v_{\text{sub.2}}$ at the lower-left corner control point of the current block are then calculated from derived motion vectors $v_{\text{sub.3}}$, $v_{\text{sub.4}}$, and $v_{\text{sub.5}}$.

[0279] It is to be noted that this method for deriving motion vector predictors may be used to derive motion vector predictors of the respective control points of the current block in Step S_{j_1} in FIG. 29 described later.

[0280] FIG. 27 is a flow chart illustrating one example of the affine merge mode.

[0281] In affine merge mode as illustrated, first, inter predictor 126 derives MV predictors of respective control points of a current block (Step S_{k_1}). The control points are an upper-left corner point of the current block and an upper-right corner point of the current block as illustrated in FIG. 25A, or an upper-left corner point of the current block, an upper-right corner point of the current block, and a lower-left corner point of the current block as illustrated in FIG. 25B.

[0282] In other words, as illustrated in FIG. 26A, inter predictor 126 checks encoded block A (left), block B (upper), block C (upper-right), block D (lower-left), and block E (upper-left) in the listed order, and identifies the first effective block encoded according to the affine mode.

[0283] When block A is identified and block A has two control points, as illustrated in FIG. 26B, inter predictor 126 calculates motion vector $v_{\text{sub.0}}$ at the upper-left corner control point of the current block and motion vector $v_{\text{sub.1}}$ at the upper-right corner control point of the current block from motion vectors $v_{\text{sub.3}}$ and $v_{\text{sub.4}}$ at the upper-left corner and the upper-right corner of the encoded block including block A. For example, inter predictor 126 calculates motion vector $v_{\text{sub.0}}$ at the upper-left corner control point of the current block and motion vector $v_{\text{sub.1}}$ at the upper-right corner control point of the current block by projecting motion vectors $v_{\text{sub.3}}$ and $v_{\text{sub.4}}$ at the upper-left corner and the upper-right corner of the encoded block onto the current block.

[0284] Alternatively, when block A is identified and block A has three control points, as illustrated in FIG. 26C, inter predictor 126 calculates motion vector $v_{\text{sub.0}}$ at the upper-left corner control point of the current block, motion vector $v_{\text{sub.1}}$ at the upper-right corner control point of the current block, and motion vector $v_{\text{sub.2}}$ at the lower-left corner control point of the current block from motion vectors $v_{\text{sub.3}}$, $v_{\text{sub.4}}$, and $v_{\text{sub.5}}$ at the upper-left corner, the upper-right corner, and the lower-left corner of the encoded block including block A. For example, inter predictor 126 calculates motion vector $v_{\text{sub.0}}$ at the upper-left corner control point of the current block, motion vector $v_{\text{sub.1}}$ at the upper-right corner control point of the current block, and motion vector $v_{\text{sub.2}}$ at the lower-left corner control point of the current block by projecting motion vectors $v_{\text{sub.3}}$, $v_{\text{sub.4}}$, and $v_{\text{sub.5}}$ at the upper-left corner, the upper-right corner, and the lower-left corner of the encoded block onto the current block.

[0285] Next, inter predictor 126 performs motion compensation of each of a plurality of sub-blocks included in the current block. In other words, inter predictor 126 calculates, for each of the plurality of sub-blocks, a motion vector of the sub-block as an affine MV, by using either (i) two motion vector predictors $v_{\text{sub.0}}$ and $v_{\text{sub.1}}$ and the expression (1A) described above or (ii) three motion

vector predictors v.sub.0, v.sub.1, and v.sub.2 and the expression (1B) described above (Step Sk_2). Inter predictor **126** then performs motion compensation of the sub-blocks using these affine MVs and encoded reference pictures (Step Sk_3). As a result, motion compensation of the current block is performed to generate a prediction image of the current block.

[MV Derivation>Affine Inter Mode]

[0286] FIG. **28A** is a conceptual diagram for illustrating an affine inter mode in which two control points are used.

[0287] In the affine inter mode, as illustrated in FIG. **28A**, a motion vector selected from motion vectors of encoded block A, block B, and block C which neighbor the current block is used as motion vector predictor v.sub.0 at the upper-left corner control point of the current block. Likewise, a motion vector selected from motion vectors of encoded block D and block E which neighbor the current block is used as motion vector predictor v.sub.1 at the upper-right corner control point of the current block.

[0288] FIG. **28B** is a conceptual diagram for illustrating an affine inter mode in which three control points are used.

[0289] In the affine inter mode, as illustrated in FIG. **28B**, a motion vector selected from motion vectors of encoded block A, block B, and block C which neighbor the current block is used as motion vector predictor v.sub.0 at the upper-left corner control point of the current block. Likewise, a motion vector selected from motion vectors of encoded block D and block E which neighbor the current block is used as motion vector predictor v.sub.1 at the upper-right corner control point of the current block. Furthermore, a motion vector selected from motion vectors of encoded block F and block G which neighbor the current block is used as motion vector predictor v.sub.2 at the lower-left corner control point of the current block.

[0290] FIG. **29** is a flow chart illustrating one example of an affine inter mode.

[0291] In the affine inter mode as illustrated, first, inter predictor **126** derives MV predictors (v.sub.0, v.sub.1) or (v.sub.0, v.sub.1, v.sub.2) of respective two or three control points of a current block (Step Sj_1). The control points are an upper-left corner point of the current block and an upper-right corner point of the current block as illustrated in FIG. **25A**, or an upper-left corner point of the current block, an upper-right corner point of the current block, and a lower-left corner point of the current block as illustrated in FIG. **25B**.

[0292] In other words, inter predictor **126** derives the motion vector predictors (v.sub.0, v.sub.1) or (v.sub.0, v.sub.1, v.sub.2) of respective two or three control points of the current block by selecting motion vectors of any of the blocks among encoded blocks in the vicinity of the respective control points of the current block illustrated in either FIG. **28A** or FIG. **28B**. At this time, inter predictor **126** encodes, in a stream, motion vector predictor selection information for identifying the selected two motion vectors.

[0293] For example, inter predictor **126** may determine, using a cost evaluation or the like, the block from which a motion vector as a motion vector predictor at a control point is selected from among encoded blocks neighboring the current block, and may describe, in a bitstream, a flag indicating which motion vector predictor has been selected.

[0294] Next, inter predictor **126** performs motion estimation (Step Sj_3 and Sj_4) while updating a motion vector predictor selected or derived in Step Sj_1 (Step Sj_2). In other words, inter predictor **126** calculates, as an affine MV, a motion vector of each of sub-blocks which corresponds to an updated motion vector predictor, using either the expression (1A) or expression (1B) described above (Step Sj_3). Inter predictor **126** then performs motion compensation of the sub-blocks using these affine MVs and encoded reference pictures (Step Sj_4). As a result, for example, inter predictor **126** determines the motion vector predictor which yields the smallest cost as the motion vector at a control point in a motion estimation loop (Step Sj_5). At this time, inter predictor **126** further encodes, in the stream, the difference value between the determined MV and the motion vector predictor as an MV difference.

[0295] Lastly, inter predictor **126** generates a prediction image for the current block by performing motion compensation of the current block using the determined MV and the encoded reference picture (Step S_{j_6}).

[MV Derivation>Affine Inter Mode]

[0296] When affine modes in which different numbers of control points (for example, two and three control points) are used may be switched and signaled at the CU level, the number of control points in an encoded block and the number of control points in a current block may be different from each other. FIG. 30A and FIG. 30B are conceptual diagrams for illustrating methods for deriving motion vector predictors at control points when the number of control points in an encoded block and the number of control points in a current block are different from each other.

[0297] For example, as illustrated in FIG. 30A, when a current block has three control points at the upper-left corner, the upper-right corner, and the lower-left corner, and block A which neighbors to the left of the current block has been encoded according to an affine mode in which two control points are used, motion vectors $V_{sub.3}$ and $V_{sub.4}$ projected at the upper-left corner position and the upper-right corner position in the encoded block including block A are derived. Motion vector predictor $v_{sub.0}$ at the upper-left corner control point of the current block and motion vector predictor $v_{sub.1}$ at the upper-right corner control point of the current block are then calculated from derived motion vectors $v_{sub.3}$ and $v_{sub.4}$. Furthermore, motion vector predictor $v_{sub.2}$ at the lower-left corner control point is calculated from derived motion vectors $v_{sub.0}$ and $v_{sub.1}$.

[0298] For example, as illustrated in FIG. 30B, when a current block has two control points at the upper-left corner and the upper-right corner, and block A which neighbors to the left of the current block has been encoded according to the affine mode in which three control points are used, motion vectors $v_{sub.3}$, $v_{sub.4}$, and $v_{sub.5}$ projected at the upper-left corner position, the upper-right corner position, and the lower-left corner position in the encoded block including block A are derived. Motion vector predictor $v_{sub.0}$ at the upper-left corner control point of the current block and motion vector predictor $v_{sub.1}$ at the upper-right corner control point of the current block are then calculated from derived motion vectors $v_{sub.3}$, $v_{sub.4}$, and $v_{sub.5}$.

[0299] It is to be noted that this method for deriving motion vector predictors may be used to derive motion vector predictors of the respective control points of the current block in Step S_{j_1} in FIG. 29.

[MV Derivation>DMVR]

[0300] FIG. 31A is a flow chart illustrating a relationship between the merge mode and DMVR.

[0301] Inter predictor **126** derives a motion vector of a current block according to the merge mode (Step S_{l_1}). Next, inter predictor **126** determines whether to perform estimation of a motion vector, that is, motion estimation (Step S_{l_2}). Here, when determining not to perform motion estimation (No in Step S_{l_2}), inter predictor **126** determines the motion vector derived in Step S_{l_1} as the final motion vector for the current block (Step S_{l_4}). In other words, in this case, the motion vector of the current block is determined according to the merge mode.

[0302] When determining to perform motion estimation in Step S_{l_1} (Yes in Step S_{l_2}), inter predictor **126** derives the final motion vector for the current block by estimating a surrounding region of the reference picture specified by the motion vector derived in Step S_{l_1} (Step S_{l_3}). In other words, in this case, the motion vector of the current block is determined according to the DMVR.

[0303] FIG. 31B is a conceptual diagram for illustrating one example of a DMVR process for determining an MV.

[0304] First, (for example, in merge mode) the best MVP which has been set to the current block is determined to be an MV candidate. A reference pixel is identified from a first reference picture (L0) which is an encoded picture in the L0 direction according to an MV candidate (L0). Likewise, a reference pixel is identified from a second reference picture (L1) which is an encoded picture in the L1 direction according to an MV candidate (L1). A template is generated by calculating an

average of these reference pixels.

[0305] Next, each of the surrounding regions of MV candidates of the first reference picture (L0) and the second reference picture (L1) are estimated, and the MV which yields the smallest cost is determined to be the final MV. It is to be noted that the cost value may be calculated, for example, using a difference value between each of the pixel values in the template and a corresponding one of the pixel values in the estimation region, the values of MV candidates, etc.

[0306] It is to be noted that the processes, configurations, and operations described here typically are basically common between the encoder and a decoder to be described later.

[0307] Exactly the same example processes described here do not always need to be performed. Any process for enabling derivation of the final MV by estimation in surrounding regions of MV candidates may be used.

[Motion Compensation>BIO/OBMC]

[0308] Motion compensation involves a mode for generating a prediction image, and correcting the prediction image. The mode is, for example, BIO and OBMC to be described later.

[0309] FIG. 32 is a flow chart illustrating one example of generation of a prediction image.

[0310] Inter predictor 126 generates a prediction image (Step Sm_1), and corrects the prediction image, for example, according to any of the modes described above (Step Sm_2).

[0311] FIG. 33 is a flow chart illustrating another example of generation of a prediction image.

[0312] Inter predictor 126 determines a motion vector of a current block (Step Sn_1). Next, inter predictor 126 generates a prediction image (Step Sn_2), and determines whether to perform a correction process (Step Sn_3). Here, when determining to perform a correction process (Yes in Step Sn_3), inter predictor 126 generates the final prediction image by correcting the prediction image (Step Sn_4). When determining not to perform a correction process (No in Step Sn_3), inter predictor 126 outputs the prediction image as the final prediction image without correcting the prediction image (Step Sn_5).

[0313] In addition, motion compensation involves a mode for correcting a luminance of a prediction image when generating the prediction image. The mode is, for example, LIC to be described later.

[0314] FIG. 34 is a flow chart illustrating another example of generation of a prediction image.

[0315] Inter predictor 126 derives a motion vector of a current block (Step So_1). Next, inter predictor 126 determines whether to perform a luminance correction process (Step So_2). Here, when determining to perform a luminance correction process (Yes in Step So_2), inter predictor 126 generates the prediction image while performing a luminance correction process (Step So_3). In other words, the prediction image is generated using LIC. When determining not to perform a luminance correction process (No in Step So_2), inter predictor 126 generates a prediction image by performing normal motion compensation without performing a luminance correction process (Step So_4).

[Motion Compensation>OBMC]

[0316] It is to be noted that an inter prediction signal may be generated using motion information for a neighboring block in addition to motion information for the current block obtained from motion estimation. More specifically, the inter prediction signal may be generated in units of a sub-block in the current block by performing a weighted addition of a prediction signal based on motion information obtained from motion estimation (in the reference picture) and a prediction signal based on motion information for a neighboring block (in the current picture). Such inter prediction (motion compensation) is also referred to as overlapped block motion compensation (OBMC).

[0317] In OBMC mode, information indicating a sub-block size for OBMC (referred to as, for example, an OBMC block size) may be signaled at the sequence level. Moreover, information indicating whether to apply the OBMC mode (referred to as, for example, an OBMC flag) may be signaled at the CU level. It is to be noted that the signaling of such information does not necessarily

need to be performed at the sequence level and CU level, and may be performed at another level (for example, at the picture level, slice level, tile level, CTU level, or sub-block level).

[0318] Examples of the OBMC mode will be described in further detail. FIGS. 35 and 36 are a flow chart and a conceptual diagram for illustrating an outline of a prediction image correction process performed by an OBMC process.

[0319] First, as illustrated in FIG. 36, a prediction image (Pred) is obtained through normal motion compensation using a motion vector (MV) assigned to the processing target (current) block. In FIG. 36, the arrow “MV” points a reference picture, and indicates what the current block of the current picture refers to in order to obtain a prediction image.

[0320] Next, a prediction image (Pred_L) is obtained by applying a motion vector (MV_L) which has been already derived for the encoded block neighboring to the left of the current block to the current block (re-using the motion vector for the current block). The motion vector (MV_L) is indicated by an arrow “MV_L” indicating a reference picture from a current block. A first correction of a prediction image is performed by overlapping two prediction images Pred and Pred_L. This provides an effect of blending the boundary between neighboring blocks.

[0321] Likewise, a prediction image (Pred_U) is obtained by applying a motion vector (MV_U) which has been already derived for the encoded block neighboring above the current block to the current block (re-using the motion vector for the current block). The motion vector (MV_U) is indicated by an arrow “MV_U” indicating a reference picture from a current block. A second correction of a prediction image is performed by overlapping the prediction image Pred_U to the prediction images (for example, Pred and Pred_L) on which the first correction has been performed. This provides an effect of blending the boundary between neighboring blocks. The prediction image obtained by the second correction is the one in which the boundary between the neighboring blocks has been blended (smoothed), and thus is the final prediction image of the current block.

[0322] Although the above example is a two-path correction method using left and upper neighboring blocks, it is to be noted that the correction method may be three- or more-path correction method using also the right neighboring block and/or the lower neighboring block.

[0323] It is to be noted that the region in which such overlapping is performed may be only part of a region near a block boundary instead of the pixel region of the entire block.

[0324] It is to be noted that the prediction image correction process according to OBMC for obtaining one prediction image Pred from one reference picture by overlapping additional prediction image Pred_L and Pred_U have been described above. However, when a prediction image is corrected based on a plurality of reference images, a similar process may be applied to each of the plurality of reference pictures. In such a case, after corrected prediction images are obtained from the respective reference pictures by performing OBMC image correction based on the plurality of reference pictures, the obtained corrected prediction images are further overlapped to obtain the final prediction image.

[0325] It is to be noted that, in OBMC, the unit of a current block may be the unit of a prediction block or the unit of a sub-block obtained by further splitting the prediction block.

[0326] One example of a method for determining whether to apply an OBMC process is a method for using an obmc_flag which is a signal indicating whether to apply an OBMC process. As one specific example, an encoder determines whether the current block belongs to a region having complicated motion. The encoder sets the obmc_flag to a value of “1” when the block belongs to a region having complicated motion and applies an OBMC process when encoding, and sets the obmc_flag to a value of “0” when the block does not belong to a region having complicated motion and encodes the block without applying an OBMC process. The decoder switches between application and non-application of an OBMC process by decoding the obmc_flag written in the stream (for example, a compressed sequence) and decoding the block by switching between the application and non-application of the OBMC process in accordance with the flag value.

[0327] Inter predictor **126** generates one rectangular prediction image for a rectangular current block in the above example. However, inter predictor **126** may generate a plurality of prediction images each having a shape different from a rectangle for the rectangular current block, and may combine the plurality of prediction images to generate the final rectangular prediction image. The shape different from a rectangle may be, for example, a triangle.

[0328] FIG. **37** is a conceptual diagram for illustrating generation of two triangular prediction images.

[0329] Inter predictor **126** generates a triangular prediction image by performing motion compensation of a first partition having a triangular shape in a current block by using a first MV of the first partition, to generate a triangular prediction image. Likewise, inter predictor **126** generates a triangular prediction image by performing motion compensation of a second partition having a triangular shape in a current block by using a second MV of the second partition, to generate a triangular prediction image. Inter predictor **126** then generates a prediction image having the same rectangular shape as the rectangular shape of the current block by combining these prediction images.

[0330] It is to be noted that, although the first partition and the second partition are triangles in the example illustrated in FIG. **37**, the first partition and the second partition may be trapezoids, or other shapes different from each other. Furthermore, although the current block includes two partitions in the example illustrated in FIG. **37**, the current block may include three or more partitions.

[0331] In addition, the first partition and the second partition may overlap with each other. In other words, the first partition and the second partition may include the same pixel region. In this case, a prediction image for a current block may be generated using a prediction image in the first partition and a prediction image in the second partition.

[0332] In addition, although an example in which a prediction image is generated for each of two partitions using inter prediction, a prediction image may be generated for at least one partition using intra prediction.

[Motion Compensation>BIO]

[0333] Next, a method for deriving a motion vector is described. First, a mode for deriving a motion vector based on a model assuming uniform linear motion will be described. This mode is also referred to as a bi-directional optical flow (BIO) mode.

[0334] FIG. **38** is a conceptual diagram for illustrating a model assuming uniform linear motion. In FIG. **38**, (vx, vy) indicates a velocity vector, and τ_0 and τ_1 indicate temporal distances between a current picture (Cur Pic) and two reference pictures (Ref0, Ref1). (MVx0, MVy0) indicate motion vectors corresponding to reference picture Ref0, and (MVx1, MVy1) indicate motion vectors corresponding to reference picture Ref1.

[0335] Here, under the assumption of uniform linear motion exhibited by velocity vectors (vx, vy), (MVx0, MVy0) and (MVx1, MVy1) are represented as (vx τ_0 , vy τ_0) and (−vx τ_1 , −vy τ_1) respectively, and the following optical flow equation (2) may be employed.

[Math. 3]

[00004]
$$\partial I^{(k)} / \partial t + v_x \partial I^{(k)} / \partial x + v_y \partial I^{(k)} / \partial y = 0. \quad (2)$$

[0336] Here, I(k) indicates a motion-compensated luma value of reference picture k (k=0, 1). This optical flow equation shows that the sum of (i) the time derivative of the luma value, (ii) the product of the horizontal velocity and the horizontal component of the spatial gradient of a reference image, and (iii) the product of the vertical velocity and the vertical component of the spatial gradient of a reference image is equal to zero. A motion vector of each block obtained from, for example, a merge list may be corrected in units of a pixel, based on a combination of the optical flow equation and Hermite interpolation.

[0337] It is to be noted that a motion vector may be derived on the decoder side using a method

other than deriving a motion vector based on a model assuming uniform linear motion. For example, a motion vector may be derived in units of a sub-block based on motion vectors of neighboring blocks.

[Motion Compensation>LIC]

[0338] Next, an example of a mode in which a prediction image (prediction) is generated by using a local illumination compensation (LIC) process will be described.

[0339] FIG. **39** is a conceptual diagram for illustrating one example of a prediction image generation method using a luminance correction process performed by a LIC process.

[0340] First, an MV is derived from an encoded reference picture, and a reference image corresponding to the current block is obtained.

[0341] Next, information indicating how the luma value changed between the reference picture and the current picture is extracted for the current block. This extraction is performed based on the luma pixel values for the encoded left neighboring reference region (surrounding reference region) and the encoded upper neighboring reference region (surrounding reference region), and the luma pixel value at the corresponding position in the reference picture specified by the derived MV. A luminance correction parameter is calculated by using the information indicating how the luma value changed.

[0342] The prediction image for the current block is generated by performing a luminance correction process in which the luminance correction parameter is applied to the reference image in the reference picture specified by the MV.

[0343] It is to be noted that the shape of the surrounding reference region illustrated in FIG. **39** is just one example; the surrounding reference region may have a different shape.

[0344] Moreover, although the process in which a prediction image is generated from a single reference picture has been described here, cases in which a prediction image is generated from a plurality of reference pictures can be described in the same manner. The prediction image may be generated after performing a luminance correction process of the reference images obtained from the reference pictures in the same manner as described above.

[0345] One example of a method for determining whether to apply a LIC process is a method for using a `lic_flag` which is a signal indicating whether to apply the LIC process. As one specific example, the encoder determines whether the current block belongs to a region having a luminance change. The encoder sets the `lic_flag` to a value of "1" when the block belongs to a region having a luminance change and applies a LIC process when encoding, and sets the `lic_flag` to a value of "0" when the block does not belong to a region having a luminance change and encodes the current block without applying a LIC process. The decoder may decode the `lic_flag` written in the stream and decode the current block by switching between application and non-application of a LIC process in accordance with the flag value.

[0346] One example of a different method of determining whether to apply a LIC process is a determining method in accordance with whether a LIC process was applied to a surrounding block. In one specific example, when the merge mode is used on the current block, whether a LIC process was applied in the encoding of the surrounding encoded block selected upon deriving the MV in the merge mode process is determined. According to the result, encoding is performed by switching between application and non-application of a LIC process. It is to be noted that, also in this example, the same processes are applied in processes at the decoder side.

[0347] An embodiment of the luminance correction (LIC) process described with reference to FIG. **39** is described in detail below.

[0348] First, inter predictor **126** derives a motion vector for obtaining a reference image corresponding to a current block to be encoded from a reference picture which is an encoded picture.

[0349] Next, inter predictor **126** extracts information indicating how the luma value of the reference picture has been changed to the luma value of the current picture, using the luma pixel

value of an encoded surrounding reference region which neighbors to the left of or above the current block and the luma value in the corresponding position in the reference picture specified by a motion vector, and calculates a luminance correction parameter. For example, it is assumed that the luma pixel value of a given pixel in the surrounding reference region in the current picture is p_0 , and that the luma pixel value of the pixel corresponding to the given pixel in the surrounding reference region in the reference picture is p_1 . Inter predictor **126** calculates coefficients A and B for optimizing $A \times p_1 + B = p_0$ as the luminance correction parameter for a plurality of pixels in the surrounding reference region.

[0350] Next, inter predictor **126** performs a luminance correction process using the luminance correction parameter for the reference image in the reference picture specified by the motion vector, to generate a prediction image for the current block. For example, it is assumed that the luma pixel value in the reference image is p_2 , and that the luminance-corrected luma pixel value of the prediction image is p_3 . Inter predictor **126** generates the prediction image after being subjected to the luminance correction process by calculating $A \times p_2 + B = p_3$ for each of the pixels in the reference image.

[0351] It is to be noted that the shape of the surrounding reference region illustrated in FIG. **39** is one example; a different shape other than the shape of the surrounding reference region may be used. In addition, part of the surrounding reference region illustrated in FIG. **39** may be used. For example, a region having a determined number of pixels extracted from each of an upper neighboring pixel and a left neighboring pixel may be used as a surrounding reference region. The determined number of pixels may be predetermined.

[0352] In addition, the surrounding reference region is not limited to a region which neighbors the current block, and may be a region which does not neighbor the current block. In the example illustrated in FIG. **39**, the surrounding reference region in the reference picture is a region specified by a motion vector in a current picture, from a surrounding reference region in the current picture. However, a region specified by another motion vector is also possible. For example, the other motion vector may be a motion vector in a surrounding reference region in the current picture.

[0353] Although operations performed by encoder **100** have been described here, it is to be noted that decoder **200** typically performs similar operations.

[0354] It is to be noted that the LIC process may be applied not only to the luma but also to chroma. At this time, a correction parameter may be derived individually for each of Y, Cb, and Cr, or a common correction parameter may be used for any of Y, Cb, and Cr.

[0355] In addition, the LIC process may be applied in units of a sub-block. For example, a correction parameter may be derived using a surrounding reference region in a current sub-block and a surrounding reference region in a reference sub-block in a reference picture specified by an MV of the current sub-block.

[Prediction Controller]

[0356] Inter predictor **128** selects one of an intra prediction signal (a signal output from intra predictor **124**) and an inter prediction signal (a signal output from inter predictor **126**), and outputs the selected signal to subtractor **104** and adder **116** as a prediction signal.

[0357] As illustrated in FIG. **1**, in various kinds of encoder examples, prediction controller **128** may output a prediction parameter which is input to entropy encoder **110**. Entropy encoder **110** may generate an encoded bitstream (or a sequence), based on the prediction parameter which is input from prediction controller **128** and quantized coefficients which are input from quantizer **108**. The prediction parameter may be used in a decoder. The decoder may receive and decode the encoded bitstream, and perform the same processes as the prediction processes performed by intra predictor **124**, inter predictor **126**, and prediction controller **128**. The prediction parameter may include (i) a selection prediction signal (for example, a motion vector, a prediction type, or a prediction mode used by intra predictor **124** or inter predictor **126**), or (ii) an optional index, a flag, or a value which is based on a prediction process performed in each of intra predictor **124**, inter

predictor **126**, and prediction controller **128**, or which indicates the prediction process.

[Mounting Example of Encoder]

[0358] FIG. **40** is a block diagram illustrating a mounting example of encoder **100**. Encoder **100** includes processor **a1** and memory **a2**. For example, the plurality of constituent elements of encoder **100** illustrated in FIG. **1** are mounted on processor **a1** and memory **a2** illustrated in FIG. **40**.

[0359] Processor **a1** is circuitry which performs information processing and is accessible to memory **a2**. For example, processor **a1** is dedicated or general electronic circuitry which encodes a video. Processor **a1** may be a processor such as a CPU. In addition, processor **a1** may be an aggregate of a plurality of electronic circuits. In addition, for example, processor **a1** may take the roles of two or more constituent elements out of the plurality of constituent elements of encoder **100** illustrated in FIG. **1**, etc.

[0360] Memory **a2** is dedicated or general memory for storing information that is used by processor **a1** to encode a video. Memory **a2** may be electronic circuitry, and may be connected to processor **a1**. In addition, memory **a2** may be included in processor **a1**. In addition, memory **a2** may be an aggregate of a plurality of electronic circuits. In addition, memory **a2** may be a magnetic disc, an optical disc, or the like, or may be represented as a storage, a recording medium, or the like. In addition, memory **a2** may be non-volatile memory, or volatile memory.

[0361] For example, memory **a2** may store a video to be encoded or a bitstream corresponding to an encoded video. In addition, memory **a2** may store a program for causing processor **a1** to encode a video.

[0362] In addition, for example, memory **a2** may take the roles of two or more constituent elements for storing information out of the plurality of constituent elements of encoder **100** illustrated in FIG. **1**, etc. For example, memory **a2** may take the roles of block memory **118** and frame memory **122** illustrated in FIG. **1**. More specifically, memory **a2** may store a reconstructed block, a reconstructed picture, etc.

[0363] It is to be noted that, in encoder **100**, all of the plurality of constituent elements indicated in FIG. **1**, etc. may not be implemented, and all the processes described above may not be performed. Part of the constituent elements indicated in FIG. **1**, etc. may be included in another device, or part of the processes described above may be performed by another device.

[Decoder]

[0364] Next, a decoder capable of decoding an encoded signal (encoded bitstream) output, for example, from encoder **100** described above will be described. FIG. **41** is a block diagram illustrating a functional configuration of decoder **200** according to an embodiment. Decoder **200** is a video decoder which decodes a video in units of a block.

[0365] As illustrated in FIG. **41**, decoder **200** includes entropy decoder **202**, inverse quantizer **204**, inverse transformer **206**, adder **208**, block memory **210**, loop filter **212**, frame memory **214**, intra predictor **216**, inter predictor **218**, and prediction controller **220**.

[0366] Decoder **200** is implemented as, for example, a generic processor and memory. In this case, when a software program stored in the memory is executed by the processor, the processor functions as entropy decoder **202**, inverse quantizer **204**, inverse transformer **206**, adder **208**, loop filter **212**, intra predictor **216**, inter predictor **218**, and prediction controller **220**. Alternatively, decoder **200** may be implemented as one or more dedicated electronic circuits corresponding to entropy decoder **202**, inverse quantizer **204**, inverse transformer **206**, adder **208**, loop filter **212**, intra predictor **216**, inter predictor **218**, and prediction controller **220**.

[0367] Hereinafter, an overall flow of processes performed by decoder **200** is described, and then each of constituent elements included in decoder **200** will be described.

[Overall Flow of Decoding Process]

[0368] FIG. **42** is a flow chart illustrating one example of an overall decoding process performed by decoder **200**.

[0369] First, entropy decoder **202** of decoder **200** identifies a splitting pattern of a block having a fixed size (for example, 128×128 pixels) (Step Sp_1). This splitting pattern is a splitting pattern selected by encoder **100**. Decoder **200** then performs processes of Step Sp_2 to Sp_6 for each of a plurality of blocks of the splitting pattern.

[0370] In other words, entropy decoder **202** decodes (specifically, entropy-decodes) encoded quantized coefficients and a prediction parameter of a current block to be decoded (also referred to as a current block) (Step Sp_2).

[0371] Next, inverse quantizer **204** performs inverse quantization of the plurality of quantized coefficients and inverse transformer **206** performs inverse transform of the result, to restore a plurality of prediction residuals (that is, a difference block) (Step Sp_3).

[0372] Next, the prediction processor including all or part of intra predictor **216**, inter predictor **218**, and prediction controller **220** generates a prediction signal (also referred to as a prediction block) of the current block (Step Sp_4).

[0373] Next, adder **208** adds the prediction block to the difference block to generate a reconstructed image (also referred to as a decoded image block) of the current block (Step Sp_5).

[0374] When the reconstructed image is generated, loop filter **212** performs filtering of the reconstructed image (Step Sp_6).

[0375] Decoder **200** then determines whether decoding of the entire picture has been finished (Step Sp_7). When determining that the decoding has not yet been finished (No in Step Sp_7), decoder **200** repeatedly executes the processes starting with Step Sp_1.

[0376] As illustrated, the processes of Steps Sp_1 to Sp_7 are performed sequentially by decoder **200**. Alternatively, two or more of the processes may be performed in parallel, the processing order of the two or more of the processes may be modified, etc.

[Entropy Decoder]

[0377] Entropy decoder **202** entropy decodes an encoded bitstream. More specifically, for example, entropy decoder **202** arithmetic decodes an encoded bitstream into a binary signal. Entropy decoder **202** then debinarizes the binary signal. With this, entropy decoder **202** outputs quantized coefficients of each block to inverse quantizer **204**. Entropy decoder **202** may output a prediction parameter included in an encoded bitstream (see FIG. 1) to intra predictor **216**, inter predictor **218**, and prediction controller **220**. Intra predictor **216**, inter predictor **218**, and prediction controller **220** in an embodiment are capable of executing the same prediction processes as those performed by intra predictor **124**, inter predictor **126**, and prediction controller **128** at the encoder side.

[Inverse Quantizer]

[0378] Inverse quantizer **204** inverse quantizes quantized coefficients of a block to be decoded (hereinafter referred to as a current block) which are inputs from entropy decoder **202**. More specifically, inverse quantizer **204** inverse quantizes quantized coefficients of the current block, based on quantization parameters corresponding to the quantized coefficients. Inverse quantizer **204** then outputs the inverse quantized transform coefficients of the current block to inverse transformer **206**.

[Inverse Transformer]

[0379] Inverse transformer **206** restores prediction errors by inverse transforming the transform coefficients which are inputs from inverse quantizer **204**.

[0380] For example, when information parsed from an encoded bitstream indicates that EMT or AMT is to be applied (for example, when an AMT flag is true), inverse transformer **206** inverse transforms the transform coefficients of the current block based on information indicating the parsed transform type.

[0381] Moreover, for example, when information parsed from an encoded bitstream indicates that NSST is to be applied, inverse transformer **206** applies a secondary inverse transform to the transform coefficients.

[Adder]

[0382] Adder **208** reconstructs the current block by adding prediction errors which are inputs from inverse transformer **206** and prediction samples which are inputs from prediction controller **220**. Adder **208** then outputs the reconstructed block to block memory **210** and loop filter **212**.

[Block Memory]

[0383] Block memory **210** is storage for storing blocks in a picture to be decoded (hereinafter referred to as a current picture) and to be referred to in intra prediction. More specifically, block memory **210** stores reconstructed blocks output from adder **208**.

[Loop Filter]

[0384] Loop filter **212** applies a loop filter to blocks reconstructed by adder **208**, and outputs the filtered reconstructed blocks to frame memory **214**, display device, etc.

[0385] When information indicating ON or OFF of an ALF parsed from an encoded bitstream indicates that an ALF is ON, one filter from among a plurality of filters is selected based on direction and activity of local gradients, and the selected filter is applied to the reconstructed block.

[Frame Memory]

[0386] Frame memory **214** is, for example, storage for storing reference pictures for use in inter prediction, and is also referred to as a frame buffer. More specifically, frame memory **214** stores a reconstructed block filtered by loop filter **212**.

[Prediction Processor (Intra Predictor, Inter Predictor, Prediction Controller)]

[0387] FIG. **43** is a flow chart illustrating one example of a process performed by a prediction processor of decoder **200**. It is to be noted that the prediction processor includes all or part of the following constituent elements: intra predictor **216**; inter predictor **218**; and prediction controller **220**.

[0388] The prediction processor generates a prediction image of a current block (Step Sq_1). This prediction image is also referred to as a prediction signal or a prediction block. It is to be noted that the prediction signal is, for example, an intra prediction signal or an inter prediction signal. Specifically, the prediction processor generates the prediction image of the current block using a reconstructed image which has been already obtained through generation of a prediction block, generation of a difference block, generation of a coefficient block, restoring of a difference block, and generation of a decoded image block.

[0389] The reconstructed image may be, for example, an image in a reference picture, or an image of a decoded block in a current picture which is the picture including the current block. The decoded block in the current picture is, for example, a neighboring block of the current block.

[0390] FIG. **44** is a flow chart illustrating another example of a process performed by the prediction processor of decoder **200**.

[0391] The prediction processor determines either a method or a mode for generating a prediction image (Step Sr_1). For example, the method or mode may be determined based on, for example, a prediction parameter, etc.

[0392] When determining a first method as a mode for generating a prediction image, the prediction processor generates a prediction image according to the first method (Step Sr_2a). When determining a second method as a mode for generating a prediction image, the prediction processor generates a prediction image according to the second method (Step Sr_2b). When determining a third method as a mode for generating a prediction image, the prediction processor generates a prediction image according to the third method (Step Sr_2c).

[0393] The first method, the second method, and the third method may be mutually different methods for generating a prediction image. Each of the first to third methods may be an inter prediction method, an intra prediction method, or another prediction method. The above-described reconstructed image may be used in these prediction methods.

[Intra Predictor]

[0394] Intra predictor **216** generates a prediction signal (intra prediction signal) by performing intra prediction by referring to a block or blocks in the current picture stored in block memory **210**,

based on the intra prediction mode parsed from the encoded bitstream. More specifically, intra predictor **216** generates an intra prediction signal by performing intra prediction by referring to samples (for example, luma and/or chroma values) of a block or blocks neighboring the current block, and then outputs the intra prediction signal to prediction controller **220**.

[0395] It is to be noted that when an intra prediction mode in which a luma block is referred to in intra prediction of a chroma block is selected, intra predictor **216** may predict the chroma component of the current block based on the luma component of the current block.

[0396] Moreover, when information parsed from an encoded bitstream indicates that PDPC is to be applied, intra predictor **216** corrects intra-predicted pixel values based on horizontal/vertical reference pixel gradients.

[Inter Predictor]

[0397] Inter predictor **218** predicts the current block by referring to a reference picture stored in frame memory **214**. Inter prediction is performed in units of a current block or a sub-block (for example, a 4×4 block) in the current block. For example, inter predictor **218** generates an inter prediction signal of the current block or the sub-block by performing motion compensation by using motion information (for example, a motion vector) parsed from an encoded bitstream (for example, a prediction parameter output from entropy decoder **202**), and outputs the inter prediction signal to prediction controller **220**.

[0398] It is to be noted that when the information parsed from the encoded bitstream indicates that the OBMC mode is to be applied, inter predictor **218** generates the inter prediction signal using motion information of a neighboring block in addition to motion information of the current block obtained from motion estimation.

[0399] Moreover, when the information parsed from the encoded bitstream indicates that the FRUC mode is to be applied, inter predictor **218** derives motion information by performing motion estimation in accordance with the pattern matching method (bilateral matching or template matching) parsed from the encoded bitstream. Inter predictor **218** then performs motion compensation (prediction) using the derived motion information.

[0400] Moreover, when the BIO mode is to be applied, inter predictor **218** derives a motion vector based on a model assuming uniform linear motion. Moreover, when the information parsed from the encoded bitstream indicates that the affine motion compensation prediction mode is to be applied, inter predictor **218** derives a motion vector of each sub-block based on motion vectors of neighboring blocks.

[MV Derivation>Normal Inter Mode]

[0401] When information parsed from an encoded bitstream indicates that the normal inter mode is to be applied, inter predictor **218** derives an MV based on the information parsed from the encoded bitstream and performs motion compensation (prediction) using the MV.

[0402] FIG. **45** is a flow chart illustrating an example of inter prediction in normal inter mode in decoder **200**.

[0403] Inter predictor **218** of decoder **200** performs motion compensation for each block. Inter predictor **218** obtains a plurality of MV candidates for a current block based on information such as MVs of a plurality of decoded blocks temporally or spatially surrounding the current block (Step Ss_1). In other words, inter predictor **218** generates an MV candidate list.

[0404] Next, inter predictor **218** extracts N (an integer of 2 or larger) MV candidates from the plurality of MV candidates obtained in Step Ss_1, as motion vector predictor candidates (also referred to as MV predictor candidates) according to a determined priority order (Step Ss_2). It is to be noted that the priority order may be determined in advance for each of the N MV predictor candidates.

[0405] Next, inter predictor **218** decodes motion vector predictor selection information from an input stream (that is, an encoded bitstream), and selects, one MV predictor candidate from the N MV predictor candidates using the decoded motion vector predictor selection information, as a

motion vector (also referred to as an MV predictor) of the current block (Step Ss_3).

[0406] Next, inter predictor **218** decodes an MV difference from the input stream, and derives an MV for a current block by adding a difference value which is the decoded MV difference and a selected motion vector predictor (Step Ss_4).

[0407] Lastly, inter predictor **218** generates a prediction image for the current block by performing motion compensation of the current block using the derived MV and the decoded reference picture (Step Ss_5).

[Prediction Controller]

[0408] Prediction controller **220** selects either the intra prediction signal or the inter prediction signal, and outputs the selected prediction signal to adder **208**. As a whole, the configurations, functions, and processes of prediction controller **220**, intra predictor **216**, and inter predictor **218** at the decoder side may correspond to the configurations, functions, and processes of prediction controller **128**, intra predictor **124**, and inter predictor **126** at the encoder side.

[Mounting Example of Decoder]

[0409] FIG. **46** is a block diagram illustrating a mounting example of decoder **200**. Decoder **200** includes processor **b1** and memory **b2**. For example, the plurality of constituent elements of decoder **200** illustrated in FIG. **41** are mounted on processor **b1** and memory **b2** illustrated in FIG. **46**.

[0410] Processor **b1** is circuitry which performs information processing and is accessible to memory **b2**. For example, processor **b1** is dedicated or general electronic circuitry which decodes a video (that is, an encoded bitstream). Processor **b1** may be a processor such as a CPU. In addition, processor **b1** may be an aggregate of a plurality of electronic circuits. In addition, for example, processor **b1** may take the roles of two or more constituent elements out of the plurality of constituent elements of decoder **200** illustrated in FIG. **41**, etc.

[0411] Memory **b2** is dedicated or general memory for storing information that is used by processor **b1** to decode an encoded bitstream. Memory **b2** may be electronic circuitry, and may be connected to processor **b1**. In addition, memory **b2** may be included in processor **b1**. In addition, memory **b2** may be an aggregate of a plurality of electronic circuits. In addition, memory **b2** may be a magnetic disc, an optical disc, or the like, or may be represented as a storage, a recording medium, or the like. In addition, memory **b2** may be a non-volatile memory, or a volatile memory.

[0412] For example, memory **b2** may store a video or a bitstream. In addition, memory **b2** may store a program for causing processor **b1** to decode an encoded bitstream.

[0413] In addition, for example, memory **b2** may take the roles of two or more constituent elements for storing information out of the plurality of constituent elements of decoder **200** illustrated in FIG. **41**, etc. Specifically, memory **b2** may take the roles of block memory **210** and frame memory **214** illustrated in FIG. **41**. More specifically, memory **b2** may store a reconstructed block, a reconstructed picture, etc.

[0414] It is to be noted that, in decoder **200**, all of the plurality of constituent elements illustrated in FIG. **41**, etc. may not be implemented, and all the processes described above may not be performed. Part of the constituent elements indicated in FIG. **41**, etc. may be included in another device, or part of the processes described above may be performed by another device.

Definitions of Terms

[0415] The respective terms may be defined as indicated below as examples.

[0416] A picture is an array of luma samples in monochrome format or an array of luma samples and two corresponding arrays of chroma samples in 4:2:0, 4:2:2, and 4:4:4 color format. A picture may be either a frame or a field.

[0417] A frame is the composition of a top field and a bottom field, where sample rows 0, 2, 4, . . . originate from the top field and sample rows 1, 3, 5, . . . originate from the bottom field.

[0418] A slice is an integer number of coding tree units contained in one independent slice segment and all subsequent dependent slice segments (if any) that precede the next independent slice

segment (if any) within the same access unit.

[0419] A tile is a rectangular region of coding tree blocks within a particular tile column and a particular tile row in a picture. A tile may be a rectangular region of the frame that is intended to be able to be decoded and encoded independently, although loop-filtering across tile edges may still be applied.

[0420] A block is an $M \times N$ (M -column by N -row) array of samples, or an $M \times N$ array of transform coefficients. A block may be a square or rectangular region of pixels including one Luma and two Chroma matrices.

[0421] A coding tree unit (CTU) may be a coding tree block of luma samples of a picture that has three sample arrays, or two corresponding coding tree blocks of chroma samples. Alternatively, a CTU may be a coding tree block of samples of one of a monochrome picture and a picture that is coded using three separate color planes and syntax structures used to code the samples.

[0422] A super block may be a square block of 64×64 pixels that consists of either 1 or 2 mode info blocks or is recursively partitioned into four 32×32 blocks, which themselves can be further partitioned.

[Aspect 1]

[0423] Aspect 1 of the present disclosure relates to indicating information essential for proper display of video within a new syntax structure. Hereinafter, information essential to the proper display of video will also be called essential display information (EDI), and a syntax structure indicating the EDI will also be called EDI syntax structure.

[0424] EDI (hereafter also called first display information) is information that is required at least for determining a pixel value to be displayed. Information essential for the proper display of video (that is, the first display information) includes, for example, information related to the image aspect ratio, information related to color description, and information related to chroma location. In addition, information essential for the proper display of video may also include field coding or frame coding for interlaced content and information regarding timing information. It should be noted that, when these two items of information are not included in first display information, these two items of information may be included in second display information which is different from the first display information. The second display information is information that is not essential for the proper display of video, and is information for realizing an additional function in the display of video such as, for example, information regarding the level of each sublayer, information regarding scalability, information regarding image quality adjustment when displaying on a display, or the like. For example, the second display information is video usability information (VUI) to be described later.

[0425] FIG. 47 illustrates an example of an essential display information (EDI) syntax structure. For example, as illustrated in FIG. 47, when all of the aforementioned 5 items of information are indicated within a video elementary stream, these information are included within the essential display information (EDI) syntax structure. At this time, the first display information may include, in this order, information related to image aspect ratio, information related to color description, and information related to chroma location. Furthermore, the information related to the image aspect ratio may include a flag indicating whether the information related to the image aspect ratio is to be encoded, the information related to color description may include a flag indicating whether the information related to color description of an image is to be encoded, and the information related to chroma location may include a flag indicating whether the information related to the chroma location of an image is to be encoded. It is sufficient that the information related to color description be encoded based only on a flag indicating whether the information related to color description is to be encoded. Through one-stage-only notification and determination which does not include a flag indicating whether or not to encode the flag indicating whether the information related to color description is to be encoded, a simpler syntax structure becomes possible.

[0426] The syntax elements in FIG. 47 have the same semantics as described in High Efficiency

Video Coding (HEVC) Annex E. The only difference is that `chroma_sample_loc_type` specifies the location of chroma samples for the whole frame in the case of progressive content, instead of the location of chroma samples for each field in the case of content in interlaced content coded with fields.

[0427] It should be noted that video usability information (VUI) syntax structure may still be present in the video elementary stream, but it does not need to contain any of the syntax elements already contained in the EDI syntax structure as illustrated in FIG. 47. For example, as illustrated in FIG. 47, when field coding or frame coding for interlaced content and information regarding timing information are included in the first display information and are indicated in the EDI syntax structure, the two pieces of information need not be included as second display information within the VUI syntax structure. On the other hand, when the two items of information are not included in the first display information, the two items of information may be included as second display information in the VUI syntax structure. Furthermore, for example, information included in the second display information may include a flag indicating whether the information is to be separately encoded. Moreover, for example, a flag indicating whether second display information is indicated in the bitstream may be written in the bitstream.

[0428] FIG. 48 illustrates an example of a video usability information (VUI) syntax structure. FIG. 48 illustrates a new version of the VUI syntax, where syntax elements have the same semantics as described in HEVC Annex E. In addition, the VUI syntax structure does not contain any syntax element already contained in the EDI syntax structure (for example, see FIG. 47), and contains information that is not essential for the proper display of video (i.e., second display information). In addition, the VUI syntax structure may contain additional information about the video (i.e., content), as long as that information is not strictly necessary for proper display of the video.

[0429] FIG. 49 illustrates an example of EDI and VUI syntax included in a sequence parameter set. As illustrated in FIG. 49, EDI and VUI syntax structures may be included into the sequence parameter set, or may be included in another similar high level syntax structure that would apply to the whole bitstream. For example, a flag “`edi_parameters_present_flag`” may signal whether the EDI syntax structure is present. In this case, for example, if the “`edi_parameters_present_flag`” is equal to 1, the EDI syntax structure is present and the VUI syntax structure may also additionally be present. Specifically, whether or not the EDI and VUI syntax structures are present may be indicated by the ON and OFF states of a flag indicating whether an EDI syntax structure is present. It should be noted that, for example, even when the flag “`edi_parameters_present_flag`” is equal to 0, if the flag “`vui_parameters_present_flag`” is equal to 1, the VUI syntax structure may be present although the EDI syntax structure is not present. Furthermore, for example, in a format in which the EDI syntax structure includes the VUI syntax structure, the flag indicating whether the VUI syntax structure is present may be signaled only when the flag indicating whether the EDI syntax structure is present is ON (i.e., is equal to 1, for example), that is, only when the EDI syntax structure is present. Furthermore, for example, in a format in which the EDI syntax structure and the VUI syntax structure are independent, the flag indicating whether the EDI syntax structure is present and the flag indicating whether the VUI syntax structure is present may be signaled separately and independently. Here, a syntax structure may mean the information represented by the syntax structure.

[0430] The main technical advantage of Aspect 1 of the present disclosure is that information essential for the proper display of video (i.e., first display information) is clearly separated from non-essential information (i.e., second display information) within the bitstream. With this, essential information has to be transmitted to the display, whereas non-essential information is intended as support for, for example, post-processing of the video (i.e., content) after decoding and before displaying. Furthermore, it may also simplify carriage of metadata in MP4 or session description protocol in streaming.

[0431] Examples of processing on the decoder-side when information essential for the proper

display of video and non-essential information are clearly separated within the bitstream will be described below with reference to the drawings. Hereinafter, information indicated within the EDI syntax structure will be called EDI information, and information indicated within the VUI syntax structure will be called VUI information.

[0432] FIG. 50 is a flowchart illustrating an example of a process of reading EDI information and VUI information by a receiver that receives a bitstream conforming to Aspect 1. Here, the receiver includes decoder 200.

[0433] As illustrated in FIG. 50, when the receiver receives the bitstream, the receiver determines whether to read EDI information from the bitstream (step S001). When the receiver determines not to read the EDI information (No in step S001), the reading process ends. In this case, the receiver creates display information necessary for displaying video on the display, by using preset default values for the information related to the image aspect ratio, the information related to color description, and the information related to chroma location included in the EDI information (not illustrated in the figure). It should be noted that, when the receiver determines not to read the EDI information in step S001, the receiver may determine whether to read VUI information.

[0434] On the other hand, when the reader determines to read the EDI information (Yes in S001), the reader reads the EDI information from the bitstream (step S002). Next, the receiver determines whether to read the VUI information (step S003). When the receiver determines not to read the VUI information (No in step S003), the reading process ends. In this case, the receiver creates display information based on the EDI information read from the bitstream (not illustrated in the figure).

[0435] On the other hand, when the receiver determines to read the VUI information (Yes in step S003), the receiver reads the VUI information from the bitstream (step S004), and the process of reading ends. In this case, the receiver creates display information based on the EDI information and VUI information read from the bitstream (not illustrated in the figure).

[0436] As described above, the receiver which includes decoder 200 may determine whether to read each of the EDI information and the VUI information, and obtain information necessary for the display of video.

[0437] Next, an example of processing up to when the EDI information and the VUI information are read from the bitstream and output to the display by the receiver will be described with reference to the drawings.

[0438] FIG. 51 is a flowchart illustrating an example of an operation of a receiver that receives a bitstream conforming to Aspect 1. In this example too, the receiver includes decoder 200.

[0439] As illustrated in FIG. 51, first, the receiver reads the EDI information from the bitstream (step S101). More specifically, the receiver reads information (i.e., EDI information) indicated within the essential display information (EDI) syntax structure which is written into the bitstream.

[0440] Next, the receiver reads the VUI information from the bitstream if the VUI information is available in the bitstream (step S102). More specifically, the receiver reads non-essential additional information (i.e., VUI information) regarding the video (i.e., content) indicated within the video usability information (VUI) syntax structure which is written into the bitstream.

[0441] Next, the receiver transmits the bitstream to decoder 200 (not illustrated in the figure). Next, decoder 200 decodes the obtained bitstream (step S103), and outputs the video obtained through the decoding to the receiver (not illustrated in the figure). The receiver obtains the video as the output of the decoding in step S103 (not illustrated in the figure).

[0442] Next, the determining unit of the receiver determines whether post-processing is applied on the video (i.e., content) (step S104). It should be noted that applying post-processing may depend on, for example, the application, user settings, display capabilities, or the content properties themselves. Furthermore, the decision may also depend on some of the information indicated within the EDI syntax structure and the VUI syntax structure.

[0443] When the determining unit determines that post-processing is to be applied to the video (Yes

in step **S104**), the receiver performs the post-processing using information read from the EDI and VUI syntax structures (step **S105**). At this time, the post-processing is applied potentially using some of the information included in the EDI syntax structure and the VUI syntax structure. Afterwards, the receiver transmits the video to the display together with the essential information for proper display indicated in the EDI syntax structure (step **S106**). Then, the process ends. [0444] On the other hand, when the determining unit determines that the post-processing is not to be applied to the video (No in step **S104**), the receiver transmits the video to the display together with the EDI information, without performing post-processing on the video (step **S106**). Then, the process ends.

[0445] FIG. **52** is a flowchart illustrating another example of an operation of a receiver that receives a bitstream conforming to Aspect 1. In this example too, the receiver includes decoder **200**.

[0446] As illustrated in FIG. **52**, first, the receiver reads the EDI information from the bitstream (step **S201**). More specifically, the receiver reads information (i.e., EDI information) indicated within the essential display information (EDI) syntax structure which is written into the bitstream. [0447] Next, the determining unit of the receiver determines whether the receiver is configured to read non-essential additional information (i.e., VUI information) regarding video (i.e., content) indicated within the video usability information (VUI) syntax structure (step **S202**).

[0448] When it is determined that the receiver is configured to read the information (VUI information) included in the video usability information (VUI) syntax structure (Yes in step **S202**), the receiver reads the VUI information from the bitstream (step **S203**). More specifically, the receiver reads non-essential additional information (i.e., VUI information) regarding the video (i.e., content) indicated in the video usability information (VUI) syntax structure, when the VUI information is available in the bitstream. Next, the receiver creates display information based on the EDI information and the VUI information (step **S204**).

[0449] On the other hand, when the receiver is not configured to read VUI information (No in step **S202**), the receiver does not read the VUI information from the bitstream (not illustrated in the figure). Next, the receiver creates display information based on the EDI information read in step **S201** (step **S205**).

[0450] Next, the receiver transmits the bitstream to decoder **200** (not illustrated in the figure). Next, decoder **200** decodes the obtained bitstream (step **S206**), and outputs the video obtained through the decoding to the receiver (not illustrated in the figure). The receiver obtains the video as the output of the decoding in step **S206** (not illustrated in the figure).

[0451] Next, the receiver transmits the decoded video to the display together with the display information created in step **S204** or step **S205** (step **S207**). Then, the process ends.

[0452] As described above, in order to create a bitstream conforming to Aspect 1, encoder **100** needs to clearly separate information that is essential for proper display of the video (i.e., first display information including, for example, information related to the image aspect ratio, color description, chroma location, field coding or frame coding of interlaced content, and timing information) from additional non-essential information (i.e., second display information). The first display information is indicated within the essential display information (EDI) syntax structure. On the other hand, the second display information is indicated within the video usability information (VUI) syntax structure.

[Variations of Aspect 1]

[0453] As a variation of Aspect 1, it may be desirable not to have to indicate the frame or field information for applications where only progressive content is intended. For that purpose, a flag “progressive_only_content_flag” may be added. This flag is equal to 1 when the syntax elements about frames and fields information are not needed, and is equal to 0 when the syntax elements about frames and fields information are needed. An example of such a syntax for the EDI parameters is illustrated in FIG. **53A** and FIG. **53B**. FIG. **53A** is a first diagram illustrating an

example of an EDI syntax structure. FIG. 53B is a second diagram illustrating an example of an EDI syntax structure.

[0454] In FIG. 53A, the flag “progressive_only_content_flag” is equal to 1, thus syntax elements about frames and fields information are not included, and a flag indicating whether information related to chroma location is present is indicated. Since the flag is equal to 1, “chroma_sample_loc_type” specifies the chroma sample location for all frames.

[0455] In other cases, for example, in the case of interlaced content, the flag “progressive_only_content_flag” is equal to 0, thus syntax elements about frames and fields are included, and a flag indicating whether information related to chroma location is present is indicated. Since the flag is equal to 1, “chroma_sample_loc_type” specifies the chroma sample location for all fields.

[0456] Furthermore, as illustrated in FIG. 53B, information related to timing information may be included within the EDI syntax structure.

[Aspect 2]

[0457] Aspect 2 of the present disclosure relates to indicating all the information (information that is essential and information that is non-essential for proper display of video) within the same data structure, but separating the essential information from the non-essential information by indicating the essential information at the beginning and indicating the non-essential information after the essential information, and further including one flag for all the non-essential information.

[0458] For example, encoder **100** may write, into the bitstream, a flag indicating whether second display information different from the first display information is indicated in the bitstream. Furthermore, for example, decoder **200** may read, from the bitstream, a flag indicating whether second display information different from the first display information is indicated in the bitstream.

[0459] An example of a data structure (i.e., syntax) conforming to Aspect 2 will be described below with reference to the drawings. FIG. 54A is a first diagram illustrating an example of syntax conforming to Aspect 2. FIG. 54B is a second diagram illustrating an example of syntax conforming to Aspect 2. FIG. 54A and FIG. 54B illustrate one possible syntax conforming to Aspect 2.

[0460] In the syntax, the flag “additional_info_present_flag” controls the presence of some additional non-essential video related information.

[0461] Next, possible exemplary operations of a receiver that receives a bitstream conforming to Aspect 2 of the present disclosure. In Aspect 2, the receiver may have operations that are similar to the operation examples of the receiver described in Aspect 1. In Aspect 1, the first display information is indicated within the EDI syntax structure and the second display information is indicated within the VUI syntax structure, whereas Aspect 2 is different in that the first display information and the second display information are indicated in the same syntax and the first display information is indicated at the beginning.

[0462] Although Aspect 2 is different from Aspect 1 in that the information in EDI information in FIG. 51 and FIG. 52 corresponds to the information indicated in the first part of the syntax structure (the part before the flag in FIG. 54A and FIG. 54B) illustrated before the “additional_info_present_flag” in FIG. 54B and the VUI information in FIG. 51 and FIG. 52 corresponds to the second part of the information indicated in the syntax structure illustrated under the “if(additional_info_present_flag)” part, these items of information are almost the same as the EDI information and the VUI information according to Aspect 1 illustrated in FIG. 51 and FIG. 52.

[0463] The main technical advantage of Aspect 2 of the present disclosure is that information essential for proper display of video (i.e., first display information) is clearly separated from non-essential information (i.e., second display information) within the bitstream. With this, essential information read from the bitstream needs to be transmitted to the display, whereas non-essential information is intended as support for, for example, post-processing of the video (i.e., content) after decoding and before displaying.

[Representative Example of Processing]

[0464] Representative examples of the processing by the above-described encoder **100** and decoder **200** will be described below.

[0465] FIG. **55** is a flowchart illustrating an operation performed by encoder **100**. Encoder **100** includes circuitry and memory coupled to the circuitry. The circuitry and the memory included in encoder **100** may correspond to processor **a1** and memory **a2** illustrated in FIG. **40**. The circuitry of encoder **100**, in operation, performs the operations described below.

[0466] For example, the circuitry of encoder **100** writes, into a bitstream, a flag indicating whether first display information, which includes information related to image aspect ratio, information related to color description, and information related to chroma location, is indicated in the bitstream (step **S301**).

[0467] Accordingly, when encoder **100** writes, into the bitstream, a flag indicating that first display information is not indicated in the bitstream, the need to search for the first display information within the bitstream is eliminated, and thus the processing amount is reduced. Therefore, the encoding efficiency of encoder **100** improves.

[0468] For example, the first display information may include, in this order, information related to image aspect ratio, information related to color description, and information related to chroma location.

[0469] Accordingly, the information related to image aspect ratio, the information related to color description, and the information related to chroma location are written into the bitstream in the order that they are included in the first display information. As a result, encoder **100** does not need to search for these items of information one after the other within the bitstream, and thus the processing amount is reduced. Therefore, the encoding efficiency of encoder **100** improves.

[0470] For example, the information related to image aspect ratio may include a flag indicating whether the information related to image aspect ratio is to be encoded, the information related to color description may include a flag indicating whether the information related to color description is to be encoded, and the information related to chroma location may include a flag indicating whether the information related to chroma location is to be encoded.

[0471] Accordingly, encoder **100** can encode the respective information according to the flag included in each of the information. As a result, all of the information need not necessarily be encoded, and thus the processing amount is reduced. Therefore, the encoding efficiency of encoder **100** improves.

[0472] For example, the circuitry of encoder **100** may write, into the bitstream, a flag indicating whether second display information different from the first display information is indicated in the bitstream (step **S302**).

[0473] Accordingly, when encoder **100** writes, into the bitstream, a flag indicating that second display information is not indicated in the bitstream, the need to search for the second display information within the bitstream is eliminated, and thus the processing amount is reduced. Therefore, the encoding efficiency of encoder **100** improves.

[0474] FIG. **56** is a flowchart illustrating an operation performed by decoder **200**. Decoder **200** includes circuitry and memory coupled to the circuitry. The circuitry and the memory included in decoder **200** may correspond to processor **b1** and memory **b2** illustrated in FIG. **46**. The circuitry of decoder **200**, in operation, performs the operations described below.

[0475] For example, the circuitry of decoder **200** reads, from a bitstream, a flag indicating whether first display information, which includes information related to image aspect ratio, information related to color description, and information related to chroma location, is indicated in the bitstream (step **S401**).

[0476] Accordingly, when decoder **200** reads, from the bitstream, a flag indicating that first display information is not indicated in the bitstream, the need to search for the first display information within the bitstream is eliminated, and thus the processing amount is reduced. Therefore, the

processing efficiency of decoder **200** improves.

[0477] For example, the first display information may include, in this order, information related to image aspect ratio, information related to color description, and information related to chroma location.

[0478] Accordingly, the information related to image aspect ratio, the information related to color description, and the information related to chroma location are read from the bitstream in the order that they are included in the first display information. As a result, decoder **200** does not need to search for these items of information one after the other within the bitstream, and thus the processing amount is reduced. Therefore, the processing efficiency of decoder **200** improves.

[0479] For example, the information related to image aspect ratio may include a flag indicating whether the information related to image aspect ratio is encoded, the information related to color description may include a flag indicating whether the information related to color description is encoded, and the information related to chroma location may include a flag indicating whether the information related to chroma location is encoded.

[0480] Accordingly, decoder **200** can decode the respective information according to the flag included in each of the information. As a result, all of the information need not necessarily be decoded, and thus the processing amount is reduced. Therefore, the processing efficiency of decoder **200** improves.

[0481] For example, the circuitry of decoder **200** may read, from the bitstream, a flag indicating whether second display information different from the first display information is indicated in the bitstream (step **S402**).

[0482] Accordingly, when decoder **200** reads, from the bitstream, a flag indicating that second display information is not indicated in the bitstream, the need to search for the second display information within the bitstream is eliminated, and thus the processing amount is reduced.

Therefore, the processing efficiency of decoder **200** improves.

[0483] Moreover, as described above, each component may be a circuit. The circuits may be integrated into a single circuit as a whole, or may be separated from each other. Moreover, each component may be implemented as a general-purpose processor, or as a dedicated processor.

[0484] Moreover, a process performed by a specific component may be performed by another component. Moreover, the order of processes may be changed, or multiple processes may be performed in parallel. Moreover, a coding device may include encoder **100** and decoder **200**.

[0485] As described above, the aspects of encoder **100** and decoder **200** have been described based on a plurality of examples, but the aspects of encoder **100** and decoder **200** are not limited to these examples. Various modifications to each of the examples that can be conceived by those skilled in the art, and forms configured by combining components in different examples without departing from the spirit of the present disclosure may be included in the scope of the aspects of encoder **100** and decoder **200**.

[Combinations Including Other Aspects]

[0486] One or more of the aspects disclosed herein may be performed by combining at least part of the other aspects in the present disclosure. In addition, one or more of the aspects disclosed herein may be performed by combining, with other aspects, part of the processes indicated in any of the flow charts according to the aspects, part of the configuration of any of the devices, part of syntaxes, etc.

[0487] It should be noted that not all of the above-described processes/components are necessarily required. A device/method may include part of the processes/components. Furthermore, the above-described processes may be executed by a decoder in the same manner as the encoder.

[Implementation and Application]

[0488] As described in each of the above embodiments, each functional or operational block may typically be realized as an MPU (micro processing unit) and memory, for example. Moreover, processes performed by each of the functional blocks may be realized as a program execution unit,

such as a processor which reads and executes software (a program) recorded on a recording medium such as ROM. The software may be distributed. The software may be recorded on a variety of recording media such as semiconductor memory. Note that each functional block can also be realized as hardware (dedicated circuit). Various combinations of hardware and software may be employed.

[0489] The processing described in each of the embodiments may be realized via integrated processing using a single apparatus (system), and, alternatively, may be realized via decentralized processing using a plurality of apparatuses. Moreover, the processor that executes the above-described program may be a single processor or a plurality of processors. In other words, integrated processing may be performed, and, alternatively, decentralized processing may be performed.

[0490] Embodiments of the present disclosure are not limited to the above exemplary embodiments; various modifications may be made to the exemplary embodiments, the results of which are also included within the scope of the embodiments of the present disclosure.

[0491] Next, application examples of the moving picture encoding method (image encoding method) and the moving picture decoding method (image decoding method) described in each of the above embodiments will be described, as well as various systems that implement the application examples. Such a system may be characterized as including an image encoder that employs the image encoding method, an image decoder that employs the image decoding method, or an image encoder-decoder that includes both the image encoder and the image decoder. Other configurations of such a system may be modified on a case-by-case basis.

Usage Examples

[0492] FIG. 57 illustrates an overall configuration of content providing system ex100 suitable for implementing a content distribution service. The area in which the communication service is provided is divided into cells of desired sizes, and base stations ex106, ex107, ex108, ex109, and ex110, which are fixed wireless stations in the illustrated example, are located in respective cells.

[0493] In content providing system ex100, devices including computer ex111, gaming device ex112, camera ex113, home appliance ex114, and smartphone ex115 are connected to internet ex101 via internet service provider ex102 or communications network ex104 and base stations ex106 through ex110. Content providing system ex100 may combine and connect any combination of the above devices. In various implementations, the devices may be directly or indirectly connected together via a telephone network or near field communication, rather than via base stations ex106 through ex110. Further, streaming server ex103 may be connected to devices including computer ex111, gaming device ex112, camera ex113, home appliance ex114, and smartphone ex115 via, for example, internet ex101. Streaming server ex103 may also be connected to, for example, a terminal in a hotspot in airplane ex117 via satellite ex116.

[0494] Note that instead of base stations ex106 through ex110, wireless access points or hotspots may be used. Streaming server ex103 may be connected to communications network ex104 directly instead of via internet ex101 or internet service provider ex102, and may be connected to airplane ex117 directly instead of via satellite ex116.

[0495] Camera ex113 is a device capable of capturing still images and video, such as a digital camera. Smartphone ex115 is a smartphone device, cellular phone, or personal handy-phone system (PHS) phone that can operate under the mobile communications system standards of the 2G, 3G, 3.9G, and 4G systems, as well as the next-generation 5G system.

[0496] Home appliance ex114 is, for example, a refrigerator or a device included in a home fuel cell cogeneration system.

[0497] In content providing system ex100, a terminal including an image and/or video capturing function is capable of, for example, live streaming by connecting to streaming server ex103 via, for example, base station ex106. When live streaming, a terminal (e.g., computer ex111, gaming device ex112, camera ex113, home appliance ex114, smartphone ex115, or a terminal in airplane ex117) may perform the encoding processing described in the above embodiments on still-image or video

content captured by a user via the terminal, may multiplex video data obtained via the encoding and audio data obtained by encoding audio corresponding to the video, and may transmit the obtained data to streaming server ex**103**. In other words, the terminal functions as the image encoder according to one aspect of the present disclosure.

[0498] Streaming server ex**103** streams transmitted content data to clients that request the stream. Client examples include computer ex**111**, gaming device ex**112**, camera ex**113**, home appliance ex**114**, smartphone ex**115**, and terminals inside airplane ex**117**, which are capable of decoding the above-described encoded data. Devices that receive the streamed data may decode and reproduce the received data. In other words, the devices may each function as the image decoder, according to one aspect of the present disclosure.

[Decentralized Processing]

[0499] Streaming server ex**103** may be realized as a plurality of servers or computers between which tasks such as the processing, recording, and streaming of data are divided. For example, streaming server ex**103** may be realized as a content delivery network (CDN) that streams content via a network connecting multiple edge servers located throughout the world. In a CDN, an edge server physically near the client may be dynamically assigned to the client. Content is cached and streamed to the edge server to reduce load times. In the event of, for example, some type of error or change in connectivity due, for example, to a spike in traffic, it is possible to stream data stably at high speeds, since it is possible to avoid affected parts of the network by, for example, dividing the processing between a plurality of edge servers, or switching the streaming duties to a different edge server and continuing streaming.

[0500] Decentralization is not limited to just the division of processing for streaming; the encoding of the captured data may be divided between and performed by the terminals, on the server side, or both. In one example, in typical encoding, the processing is performed in two loops. The first loop is for detecting how complicated the image is on a frame-by-frame or scene-by-scene basis, or detecting the encoding load. The second loop is for processing that maintains image quality and improves encoding efficiency. For example, it is possible to reduce the processing load of the terminals and improve the quality and encoding efficiency of the content by having the terminals perform the first loop of the encoding and having the server side that received the content perform the second loop of the encoding. In such a case, upon receipt of a decoding request, it is possible for the encoded data resulting from the first loop performed by one terminal to be received and reproduced on another terminal in approximately real time. This makes it possible to realize smooth, real-time streaming.

[0501] In another example, camera ex**113** or the like extracts a feature amount (an amount of features or characteristics) from an image, compresses data related to the feature amount as metadata, and transmits the compressed metadata to a server. For example, the server determines the significance of an object based on the feature amount and changes the quantization accuracy accordingly to perform compression suitable for the meaning (or content significance) of the image. Feature amount data is particularly effective in improving the precision and efficiency of motion vector prediction during the second compression pass performed by the server. Moreover, encoding that has a relatively low processing load, such as variable length coding (VLC), may be handled by the terminal, and encoding that has a relatively high processing load, such as context-adaptive binary arithmetic coding (CABAC), may be handled by the server.

[0502] In yet another example, there are instances in which a plurality of videos of approximately the same scene are captured by a plurality of terminals in, for example, a stadium, shopping mall, or factory. In such a case, for example, the encoding may be decentralized by dividing processing tasks between the plurality of terminals that captured the videos and, if necessary, other terminals that did not capture the videos, and the server, on a per-unit basis. The units may be, for example, groups of pictures (GOP), pictures, or tiles resulting from dividing a picture. This makes it possible to reduce load times and achieve streaming that is closer to real time.

[0503] Since the videos are of approximately the same scene, management and/or instructions may be carried out by the server so that the videos captured by the terminals can be cross-referenced. Moreover, the server may receive encoded data from the terminals, change the reference relationship between items of data, or correct or replace pictures themselves, and then perform the encoding. This makes it possible to generate a stream with increased quality and efficiency for the individual items of data.

[0504] Furthermore, the server may stream video data after performing transcoding to convert the encoding format of the video data. For example, the server may convert the encoding format from MPEG to VP (e.g., VP9), may convert H.264 to H.265, etc.

[0505] In this way, encoding can be performed by a terminal or one or more servers. Accordingly, although the device that performs the encoding is referred to as a “server” or “terminal” in the following description, some or all of the processes performed by the server may be performed by the terminal, and likewise some or all of the processes performed by the terminal may be performed by the server. This also applies to decoding processes.

[3D, Multi-Angle]

[0506] There has been an increase in usage of images or videos combined from images or videos of different scenes concurrently captured, or of the same scene captured from different angles, by a plurality of terminals such as camera ex113 and/or smartphone ex115. Videos captured by the terminals may be combined based on, for example, the separately obtained relative positional relationship between the terminals, or regions in a video having matching feature points.

[0507] In addition to the encoding of two-dimensional moving pictures, the server may encode a still image based on scene analysis of a moving picture, either automatically or at a point in time specified by the user, and transmit the encoded still image to a reception terminal. Furthermore, when the server can obtain the relative positional relationship between the video capturing terminals, in addition to two-dimensional moving pictures, the server can generate three-dimensional geometry of a scene based on video of the same scene captured from different angles. The server may separately encode three-dimensional data generated from, for example, a point cloud and, based on a result of recognizing or tracking a person or object using three-dimensional data, may select or reconstruct and generate a video to be transmitted to a reception terminal, from videos captured by a plurality of terminals.

[0508] This allows the user to enjoy a scene by freely selecting videos corresponding to the video capturing terminals, and allows the user to enjoy the content obtained by extracting a video at a selected viewpoint from three-dimensional data reconstructed from a plurality of images or videos. Furthermore, as with video, sound may be recorded from relatively different angles, and the server may multiplex audio from a specific angle or space with the corresponding video, and transmit the multiplexed video and audio.

[0509] In recent years, content that is a composite of the real world and a virtual world, such as virtual reality (VR) and augmented reality (AR) content, has also become popular. In the case of VR images, the server may create images from the viewpoints of both the left and right eyes, and perform encoding that tolerates reference between the two viewpoint images, such as multi-view coding (MVC), and, alternatively, may encode the images as separate streams without referencing. When the images are decoded as separate streams, the streams may be synchronized when reproduced, so as to recreate a virtual three-dimensional space in accordance with the viewpoint of the user.

[0510] In the case of AR images, the server may superimpose virtual object information existing in a virtual space onto camera information representing a real-world space, based on a three-dimensional position or movement from the perspective of the user. The decoder may obtain or store virtual object information and three-dimensional data, generate two-dimensional images based on movement from the perspective of the user, and then generate superimposed data by seamlessly connecting the images. Alternatively, the decoder may transmit, to the server, motion

from the perspective of the user in addition to a request for virtual object information. The server may generate superimposed data based on three-dimensional data stored in the server in accordance with the received motion, and encode and stream the generated superimposed data to the decoder. Note that superimposed data typically includes, in addition to RGB values, an a value indicating transparency, and the server sets the a value for sections other than the object generated from three-dimensional data to, for example, 0, and may perform the encoding while those sections are transparent. Alternatively, the server may set the background to a determined RGB value, such as a chroma key, and generate data in which areas other than the object are set as the background. The determined RGB value may be predetermined.

[0511] Decoding of similarly streamed data may be performed by the client (e.g., the terminals), on the server side, or divided therebetween. In one example, one terminal may transmit a reception request to a server, the requested content may be received and decoded by another terminal, and a decoded signal may be transmitted to a device having a display. It is possible to reproduce high image quality data by decentralizing processing and appropriately selecting content regardless of the processing ability of the communications terminal itself. In yet another example, while a TV, for example, is receiving image data that is large in size, a region of a picture, such as a tile obtained by dividing the picture, may be decoded and displayed on a personal terminal or terminals of a viewer or viewers of the TV. This makes it possible for the viewers to share a big-picture view as well as for each viewer to check his or her assigned area, or inspect a region in further detail up close.

[0512] In situations in which a plurality of wireless connections are possible over near, mid, and far distances, indoors or outdoors, it may be possible to seamlessly receive content using a streaming system standard such as MPEG-DASH. The user may switch between data in real time while freely selecting a decoder or display apparatus including the user's terminal, displays arranged indoors or outdoors, etc. Moreover, using, for example, information on the position of the user, decoding can be performed while switching which terminal handles decoding and which terminal handles the displaying of content. This makes it possible to map and display information, while the user is on the move in route to a destination, on the wall of a nearby building in which a device capable of displaying content is embedded, or on part of the ground. Moreover, it is also possible to switch the bit rate of the received data based on the accessibility to the encoded data on a network, such as when encoded data is cached on a server quickly accessible from the reception terminal, or when encoded data is copied to an edge server in a content delivery service.

[Scalable Encoding]

[0513] The switching of content will be described with reference to a scalable stream, illustrated in FIG. 58, which is compression coded via implementation of the moving picture encoding method described in the above embodiments. The server may have a configuration in which content is switched while making use of the temporal and/or spatial scalability of a stream, which is achieved by division into and encoding of layers, as illustrated in the figure. Note that there may be a plurality of individual streams that are of the same content but different quality. In other words, by determining which layer to decode based on internal factors, such as the processing ability on the decoder side, and external factors, such as communication bandwidth, the decoder side can freely switch between low resolution content and high resolution content while decoding. For example, in a case in which the user wants to continue watching, for example at home on a device such as a TV connected to the internet, a video that the user had been previously watching on smartphone ex115 while on the move, the device can simply decode the same stream up to a different layer, which reduces the server side load.

[0514] Furthermore, in addition to the configuration described above, in which scalability is achieved as a result of the pictures being encoded per layer, with the enhancement layer being above the base layer, the enhancement layer may include metadata based on, for example, statistical information on the image. The decoder side may generate high image quality content by

performing super-resolution imaging on a picture in the base layer based on the metadata. Super-resolution imaging may improve the SN ratio while maintaining resolution and/or increasing resolution. Metadata includes information for identifying a linear or a non-linear filter coefficient, as used in super-resolution processing, or information identifying a parameter value in filter processing, machine learning, or a least squares method used in super-resolution processing. [0515] Alternatively, a configuration may be provided in which a picture is divided into, for example, tiles in accordance with, for example, the meaning of an object in the image. On the decoder side, only a partial region is decoded by selecting a tile to decode. Further, by storing an attribute of the object (person, car, ball, etc.) and a position of the object in the video (coordinates in identical images) as metadata, the decoder side can identify the position of a desired object based on the metadata and determine which tile or tiles include that object. For example, as illustrated in FIG. 59, metadata may be stored using a data storage structure different from pixel data, such as an SEI (supplemental enhancement information) message in HEVC. This metadata indicates, for example, the position, size, or color of the main object.

[0516] Metadata may be stored in units of a plurality of pictures, such as stream, sequence, or random access units. The decoder side can obtain, for example, the time at which a specific person appears in the video, and by fitting the time information with picture unit information, can identify a picture in which the object is present, and can determine the position of the object in the picture.

[Web Page Optimization]

[0517] FIG. 60 illustrates an example of a display screen of a web page on computer ex111, for example. FIG. 61 illustrates an example of a display screen of a web page on smartphone ex115, for example. As illustrated in FIG. 60 and FIG. 61, a web page may include a plurality of image links that are links to image content, and the appearance of the web page may differ depending on the device used to view the web page. When a plurality of image links are viewable on the screen, until the user explicitly selects an image link, or until the image link is in the approximate center of the screen or the entire image link fits in the screen, the display apparatus (decoder) may display, as the image links, still images included in the content or I pictures; may display video such as an animated gif using a plurality of still images or I pictures; or may receive only the base layer, and decode and display the video.

[0518] When an image link is selected by the user, the display apparatus performs decoding while, for example, giving the highest priority to the base layer. Note that if there is information in the HTML code of the web page indicating that the content is scalable, the display apparatus may decode up to the enhancement layer. Further, in order to guarantee real-time reproduction, before a selection is made or when the bandwidth is severely limited, the display apparatus can reduce delay between the point in time at which the leading picture is decoded and the point in time at which the decoded picture is displayed (that is, the delay between the start of the decoding of the content to the displaying of the content) by decoding and displaying only forward reference pictures (I picture, P picture, forward reference B picture). Still further, the display apparatus may purposely ignore the reference relationship between pictures, and coarsely decode all B and P pictures as forward reference pictures, and then perform normal decoding as the number of pictures received over time increases.

[Autonomous Driving]

[0519] When transmitting and receiving still image or video data such as two- or three-dimensional map information for autonomous driving or assisted driving of an automobile, the reception terminal may receive, in addition to image data belonging to one or more layers, information on, for example, the weather or road construction as metadata, and associate the metadata with the image data upon decoding. Note that metadata may be assigned per layer and, alternatively, may simply be multiplexed with the image data.

[0520] In such a case, since the automobile, drone, airplane, etc., containing the reception terminal is mobile, the reception terminal may seamlessly receive and perform decoding while switching

between base stations among base stations ex106 through ex110 by transmitting information indicating the position of the reception terminal. Moreover, in accordance with the selection made by the user, the situation of the user, and/or the bandwidth of the connection, the reception terminal may dynamically select to what extent the metadata is received, or to what extent the map information, for example, is updated.

[0521] In content providing system ex100, the client may receive, decode, and reproduce, in real time, encoded information transmitted by the user.

[Streaming of Individual Content]

[0522] In content providing system ex100, in addition to high image quality, long content distributed by a video distribution entity, unicast or multicast streaming of low image quality, and short content from an individual are also possible. Such content from individuals is likely to further increase in popularity. The server may first perform editing processing on the content before the encoding processing, in order to refine the individual content. This may be achieved using the following configuration, for example.

[0523] In real time while capturing video or image content, or after the content has been captured and accumulated, the server performs recognition processing based on the raw data or encoded data, such as capture error processing, scene search processing, meaning analysis, and/or object detection processing. Then, based on the result of the recognition processing, the server—either when prompted or automatically—edits the content, examples of which include: correction such as focus and/or motion blur correction; removing low-priority scenes such as scenes that are low in brightness compared to other pictures, or out of focus; object edge adjustment; and color tone adjustment. The server encodes the edited data based on the result of the editing. It is known that excessively long videos tend to receive fewer views. Accordingly, in order to keep the content within a specific length that scales with the length of the original video, the server may, in addition to the low-priority scenes described above, automatically clip out scenes with low movement, based on an image processing result. Alternatively, the server may generate and encode a video digest based on a result of an analysis of the meaning of a scene.

[0524] There may be instances in which individual content may include content that infringes a copyright, moral right, portrait rights, etc. Such instance may lead to an unfavorable situation for the creator, such as when content is shared beyond the scope intended by the creator. Accordingly, before encoding, the server may, for example, edit images so as to blur faces of people in the periphery of the screen or blur the inside of a house, for example. Further, the server may be configured to recognize the faces of people other than a registered person in images to be encoded, and when such faces appear in an image, may apply a mosaic filter, for example, to the face of the person. Alternatively, as pre- or post-processing for encoding, the user may specify, for copyright reasons, a region of an image including a person or a region of the background to be processed. The server may process the specified region by, for example, replacing the region with a different image, or blurring the region. If the region includes a person, the person may be tracked in the moving picture, and the person's head region may be replaced with another image as the person moves.

[0525] Since there is a demand for real-time viewing of content produced by individuals, which tends to be small in data size, the decoder may first receive the base layer as the highest priority, and perform decoding and reproduction, although this may differ depending on bandwidth. When the content is reproduced two or more times, such as when the decoder receives the enhancement layer during decoding and reproduction of the base layer, and loops the reproduction, the decoder may reproduce a high image quality video including the enhancement layer. If the stream is encoded using such scalable encoding, the video may be low quality when in an unselected state or at the start of the video, but it can offer an experience in which the image quality of the stream progressively increases in an intelligent manner. This is not limited to just scalable encoding; the same experience can be offered by configuring a single stream from a low quality stream

reproduced for the first time and a second stream encoded using the first stream as a reference.

Other Implementation and Application Examples

[0526] The encoding and decoding may be performed by LSI (large scale integration circuitry) **ex500** (see FIG. 57), which is typically included in each terminal. LSI **ex500** may be configured of a single chip or a plurality of chips. Software for encoding and decoding moving pictures may be integrated into some type of a recording medium (such as a CD-ROM, a flexible disk, or a hard disk) that is readable by, for example, computer **ex111**, and the encoding and decoding may be performed using the software. Furthermore, when smartphone **ex115** is equipped with a camera, the video data obtained by the camera may be transmitted. In this case, the video data may be coded by LSI **ex500** included in smartphone **ex115**.

[0527] Note that LSI **ex500** may be configured to download and activate an application. In such a case, the terminal first determines whether it is compatible with the scheme used to encode the content, or whether it is capable of executing a specific service. When the terminal is not compatible with the encoding scheme of the content, or when the terminal is not capable of executing a specific service, the terminal may first download a codec or application software and then obtain and reproduce the content.

[0528] Aside from the example of content providing system **ex100** that uses internet **ex101**, at least the moving picture encoder (image encoder) or the moving picture decoder (image decoder) described in the above embodiments may be implemented in a digital broadcasting system. The same encoding processing and decoding processing may be applied to transmit and receive broadcast radio waves superimposed with multiplexed audio and video data using, for example, a satellite, even though this is geared toward multicast, whereas unicast is easier with content providing system **ex100**.

[Hardware Configuration]

[0529] FIG. 62 illustrates further details of smartphone **ex115** shown in FIG. 57. FIG. 63 illustrates a configuration example of smartphone **ex115**. Smartphone **ex115** includes antenna **ex450** for transmitting and receiving radio waves to and from base station **ex110**, camera **ex465** capable of capturing video and still images, and display **ex458** that displays decoded data, such as video captured by camera **ex465** and video received by antenna **ex450**. Smartphone **ex115** further includes user interface **ex466** such as a touch panel, audio output unit **ex457** such as a speaker for outputting speech or other audio, audio input unit **ex456** such as a microphone for audio input, memory **ex467** capable of storing decoded data such as captured video or still images, recorded audio, received video or still images, and mail, as well as decoded data, and slot **ex464** which is an interface for SIM **ex468** for authorizing access to a network and various data. Note that external memory may be used instead of memory **ex467**.

[0530] Main controller **ex460**, which may comprehensively control display **ex458** and user interface **ex466**, power supply circuit **ex461**, user interface input controller **ex462**, video signal processor **ex455**, camera interface **ex463**, display controller **ex459**, modulator/demodulator **ex452**, multiplexer/demultiplexer **ex453**, audio signal processor **ex454**, slot **ex464**, and memory **ex467** are connected via bus **ex470**.

[0531] When the user turns on the power button of power supply circuit **ex461**, smartphone **ex115** is powered on into an operable state, and each component is supplied with power from a battery pack.

[0532] Smartphone **ex115** performs processing for, for example, calling and data transmission, based on control performed by main controller **ex460**, which includes a CPU, ROM, and RAM. When making calls, an audio signal recorded by audio input unit **ex456** is converted into a digital audio signal by audio signal processor **ex454**, to which spread spectrum processing is applied by modulator/demodulator **ex452** and digital-analog conversion, and frequency conversion processing is applied by transmitter/receiver **ex451**, and the resulting signal is transmitted via antenna **ex450**. The received data is amplified, frequency converted, and analog-digital converted, inverse spread

spectrum processed by modulator/demodulator ex452, converted into an analog audio signal by audio signal processor ex454, and then output from audio output unit ex457. In data transmission mode, text, still-image, or video data may be transmitted under control of main controller ex460 via user interface input controller ex462 based on operation of user interface ex466 of the main body, for example. Similar transmission and reception processing is performed. In data transmission mode, when sending a video, still image, or video and audio, video signal processor ex455 compression encodes, via the moving picture encoding method described in the above embodiments, a video signal stored in memory ex467 or a video signal input from camera ex465, and transmits the encoded video data to multiplexer/demultiplexer ex453. Audio signal processor ex454 encodes an audio signal recorded by audio input unit ex456 while camera ex465 is capturing a video or still image, and transmits the encoded audio data to multiplexer/demultiplexer ex453. Multiplexer/demultiplexer ex453 multiplexes the encoded video data and encoded audio data using a determined scheme, modulates and converts the data using modulator/demodulator (modulator/demodulator circuit) ex452 and transmitter/receiver ex451, and transmits the result via antenna ex450. The determined scheme may be predetermined.

[0533] When video appended in an email or a chat, or a video linked from a web page, is received, for example, in order to decode the multiplexed data received via antenna ex450, multiplexer/demultiplexer ex453 demultiplexes the multiplexed data to divide the multiplexed data into a bitstream of video data and a bitstream of audio data, supplies the encoded video data to video signal processor ex455 via synchronous bus ex470, and supplies the encoded audio data to audio signal processor ex454 via synchronous bus ex470. Video signal processor ex455 decodes the video signal using a moving picture decoding method corresponding to the moving picture encoding method described in the above embodiments, and video or a still image included in the linked moving picture file is displayed on display ex458 via display controller ex459. Audio signal processor ex454 decodes the audio signal and outputs audio from audio output unit ex457. Since real-time streaming is becoming increasingly popular, there may be instances in which reproduction of the audio may be socially inappropriate, depending on the user's environment. Accordingly, as an initial value, a configuration in which only video data is reproduced, i.e., the audio signal is not reproduced, may be preferable; audio may be synchronized and reproduced only when an input, such as when the user clicks video data, is received.

[0534] Although smartphone ex115 was used in the above example, other implementations are conceivable: a transceiver terminal including both an encoder and a decoder; a transmitter terminal including only an encoder; and a receiver terminal including only a decoder. In the description of the digital broadcasting system, an example is given in which multiplexed data obtained as a result of video data being multiplexed with audio data is received or transmitted. The multiplexed data, however, may be video data multiplexed with data other than audio data, such as text data related to the video. Further, the video data itself rather than multiplexed data may be received or transmitted.

[0535] Although main controller ex460 including a CPU is described as controlling the encoding or decoding processes, various terminals often include GPUs. Accordingly, a configuration is acceptable in which a large area is processed at once by making use of the performance ability of the GPU via memory shared by the CPU and GPU, or memory including an address that is managed so as to allow common usage by the CPU and GPU. This makes it possible to shorten encoding time, maintain the real-time nature of the stream, and reduce delay. In particular, processing relating to motion estimation, deblocking filtering, sample adaptive offset (SAO), and transformation/quantization can be effectively carried out by the GPU instead of the CPU in units of pictures, for example, all at once.

[0536] Although only some exemplary embodiments of the present disclosure have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the present disclosure. Accordingly, all such modifications are intended to be

included within the scope of the present disclosure.

INDUSTRIAL APPLICABILITY

[0537] The present disclosure is applicable to a television receiver, a digital video recorder, a car navigation, a mobile phone, a digital camera, a digital video camera, a teleconference system, an electronic mirror, etc.

Claims

1. An encoder, comprising: circuitry; and memory coupled to the circuitry, wherein in operation, the circuitry: writes, into a sequence parameter set (SPS) which is header information of a picture sequence including pictures, video usability information (VUI) syntax; and writes first syntax different from the VUI syntax into the SPS before the VUI syntax, the first syntax including at least one parameter related to display timings of the pictures included in the picture sequence, and wherein the circuitry writes, into the SPS, a flag indicating whether the first syntax is provided in the SPS, and the at least one parameter includes a parameter indicating a number of time units per 1 clock tick.
 2. A decoder, comprising: circuitry; and memory coupled to the circuitry, wherein in operation, the circuitry: reads, from a sequence parameter set (SPS) which is header information of a picture sequence including pictures, video usability information (VUI) syntax; and reads first syntax different from the VUI syntax from the SPS before the VUI syntax, the first syntax including at least one parameter related to display timings of the pictures included in the picture sequence, and wherein the circuitry reads, from the SPS, a flag indicating whether the first syntax is provided in the SPS, and the at least one parameter includes a parameter indicating a number of time units per 1 clock tick.
 3. A bitstream generation device, comprising: circuitry; and memory coupled to the circuitry, wherein in operation, the circuitry: writes, into a sequence parameter set (SPS) which is header information of a picture sequence including pictures, video usability information (VUI) syntax; writes first syntax different from the VUI syntax into the SPS before the VUI syntax, the first syntax including at least one parameter related to display timings of the pictures included in the picture sequence; outputs a bitstream including the SPS, and wherein the circuitry writes, into the SPS, a flag indicating whether the first syntax is provided in the SPS, and the at least one parameter includes a parameter indicating a number of time units per 1 clock tick.
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